



**REPUBLIC OF KENYA**

**COMPETENCY BASED MODULAR CURRICULUM**

**FOR**

**COMPUTER SCIENCE**

**KNQF LEVEL 6**

**(CYCLE 3)**

**PROGRAMME ISCED CODE: 0613 554 A.**



**TVET CDACC**

**P.O. BOX 15745-00100**

**NAIROBI**

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## **FOREWORD**

The provision of quality education and training is fundamental to the Government's overall strategy for social and economic development. Quality education and training contribute to the achievement of Kenya's development blueprint and sustainable development goals.

Reforms in the education sector are necessary to achieve Kenya Vision 2030 and meet the provisions of the Constitution of Kenya 2010. The education sector had to be aligned to the Constitution, and this resulted in the formulation of the Policy Framework for Reforming Education and Training in Kenya (Sessional Paper No. 14 of 2012). A key feature of this policy is the radical change in the design and delivery of TVET training. This policy document requires that training in TVET be competency-based, curriculum development be industry-led, certification be based on demonstration of competence, and the mode of delivery allow for multiple entry and exit in TVET programmes.

These reforms demand that Industry takes a leading role in curriculum development to ensure the curriculum addresses its competence needs. It is against this background that this curriculum has been developed. For trainees to build their skills on foundational hands-on activities of the occupation, units of learning are grouped in modules. This has eliminated duplication of content and streamlined exemptions based on skills acquired as a trainee progresses in the up-skilling process, while at the same time allowing trainees to be employable in the shortest time possible through the acquisition of part qualifications.

It is my conviction that this curriculum will play a great role in developing competent human resources for the Computer Science Sector's growth and development.

**PRINCIPAL SECRETARY**

**STATE DEPARTMENT FOR TVET**

**MINISTRY OF EDUCATION**

## **PREFACE**

Kenya Vision 2030 aims to transform Kenya into a newly industrializing middle-income country, providing high-quality life to all its citizens by the year 2030. Kenya intends to create globally competitive and adaptive human resource base to meet the requirements of a rapidly industrializing economy through lifelong education and training. TVET has a responsibility to facilitate the process of inculcating knowledge, skills, and worker behaviour necessary for catapulting the nation to a globally competitive country, hence the paradigm shift to embrace Competency-Based Education and Training (CBET).

CAP 210A and Sessional Paper No. 1 of 2019 on Reforming Education and Training in Kenya for Sustainable Development emphasized the need to reform curriculum development, assessment, and certification. This called for a shift to CBET to address the mismatch between skills acquired through training and skills needed by industry, as well as increase the global competitiveness of the Kenyan labour force.

This curriculum has been developed in adherence to the Kenya National Qualifications Framework and CBETA standards and guidelines. The curriculum is designed and organized into Units of Learning with Learning Outcomes, suggested delivery methods, learning resources, and methods of assessing the trainee's achievement. In addition, the units of learning have been grouped in modules to concretize the skills acquisition process and streamline upskilling.

I am grateful to all expert trainers and everyone who played a role in translating the Occupational Standards into this competency-based modular curriculum.

**CHAIRMAN**  
**TVET CDACC**

## **ACKNOWLEDGMENT**

This curriculum has been designed for competency-based training and has independent units of learning that allow the trainee flexibility in entry and exit. In developing the curriculum, significant involvement and support were received from expert trainers, institutions and organizations.

I recognize with appreciation the role of the ICT National Sector Skills Committee (NSSC) in ensuring that competencies required by the industry are addressed in the curriculum. I also thank all stakeholders in the ICT sector for their valuable input and everyone who participated in developing this curriculum.

I am convinced that this curriculum will go a long way in ensuring that individuals aspiring to work in the ICT Sector acquire competencies to perform their work more efficiently and effectively.

**CEO/ COUNCIL SECRETARY**

**TVET CDACC**

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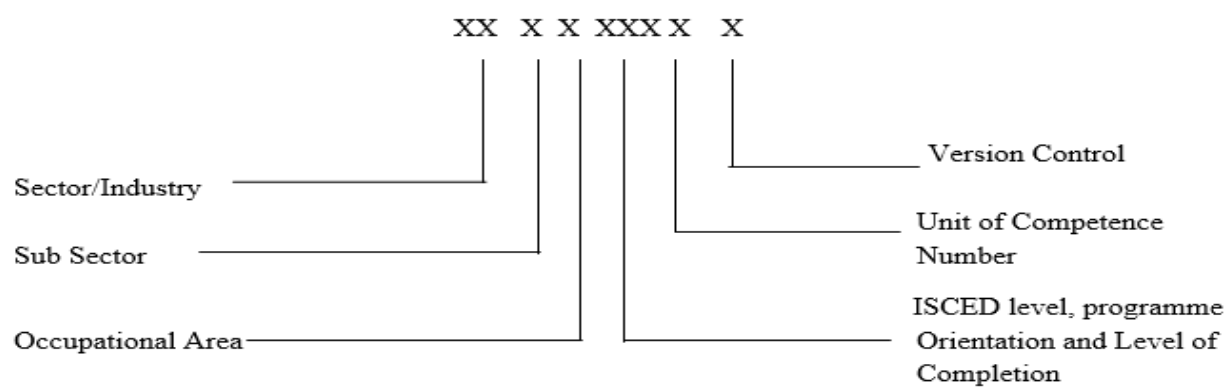
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## **ABBREVIATIONS AND ACRONYMS**

|        |                                                             |
|--------|-------------------------------------------------------------|
| AIDS   | Acquired Immunodeficiency Syndrome                          |
| BC     | Basic Unit                                                  |
| CBET   | Competency Based Education and Training                     |
| CC     | Common unit                                                 |
| CDACC  | Curriculum Development Assessment Certification Council     |
| CEO    | Council Secretary                                           |
| CR     | Core Unit                                                   |
| CU     | Curriculum                                                  |
| HIV    | Human Immuno-Deficiency Virus                               |
| KCSE   | Kenya Certificate of Secondary Education                    |
| KNQA   | Kenya National Qualifications Authority                     |
| LCD    | Liquid Crystal Display                                      |
| MA     | Control version                                             |
| OSH    | Occupational Safety and Health                              |
| PESTEL | Political Environmental Social Technological Economic Legal |
| PPE    | Personal Protective Equipment                               |
| Q&A    | Questions and Answer                                        |
| SSAC   | Sector Skills Advisory Committee                            |
| SWOT   | Strength Weakness Opportunity Threat                        |
| TVET   | Technical and Vocational Education and Training             |

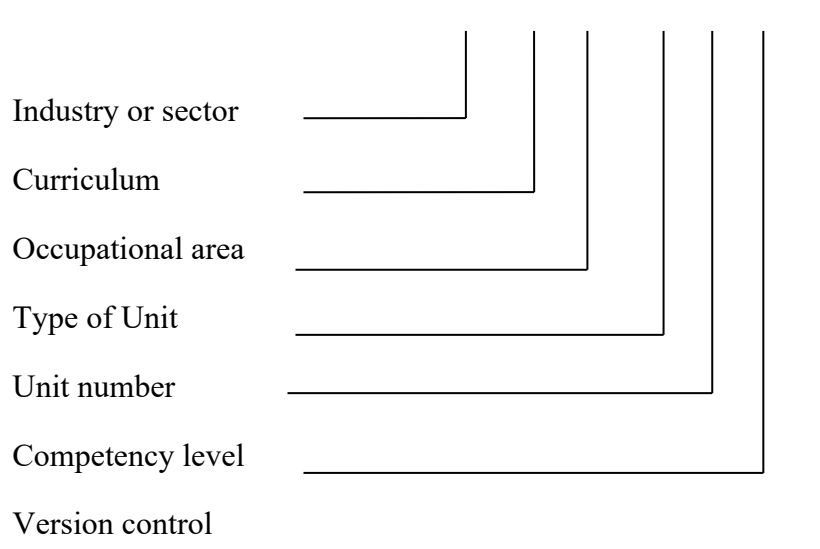
## KEY TO UNIT CODE

### KEY TO ISCED UNIT CODE



### KEY TO TVET CDACC UNIT CODE

ICT /OS /CS /CR / 06 / 6 /MA





## COURSE OVERVIEW

Computer Science Level 6 qualification consists of competencies that an individual must achieve to offer computer science services. It involves Computer organization and architecture, operating systems, Mathematics for computer science, Fundamentals of programming, Database management skills, Information system, Networking and distributed systems, Artificial intelligence, Algorithms and data structures, Web design skills and Graphic design

### Summary of Units of Learning

| ISCED<br>Unit<br>Code | TVE CDACC<br>Unit<br>Code | Unit of Learning Title                 | Duration<br>in<br>Hours | Credit<br>Factor |
|-----------------------|---------------------------|----------------------------------------|-------------------------|------------------|
| <b>MODULE I</b>       |                           |                                        |                         |                  |
| 0613 554<br>01A       | ICT/CU/CS/CR/01/6/MA      | Computer Organisation and Architecture | 180                     | 18.0             |
| 0613 554<br>02A       | ICT/CU/CS/CR/02/6/MA      | Operating Systems Configuration        | 240                     | 24.0             |
|                       |                           | <b>SUB TOTAL</b>                       | <b>420</b>              | <b>42.0</b>      |
| <b>MODULE II</b>      |                           |                                        |                         |                  |
| 0613 554<br>03A       | ICT/CU/CS/CR/03/6/MA      | Networking and Distributed Systems     | 220                     | 22.0             |
| 0613 554<br>04A       | ICT/CU/CS/CR/04/6/MA      | Graphics Design                        | 190                     | 19.0             |
|                       |                           | <b>SUB TOTAL</b>                       | <b>410</b>              | <b>41.0</b>      |
| <b>MODULE III</b>     |                           |                                        |                         |                  |
| 0613 554<br>05A       | ICT/CU/CS/CR/05/6/MA      | Database Management                    | 210                     | 21.0             |
| 0613 554<br>06A       | ICT/CU/CS/CR/06/6/MA      | Web Designing                          | 220                     | 22.0             |
|                       |                           | <b>SUB TOTAL</b>                       | <b>430</b>              | <b>43.0</b>      |

| <b>MODULE IV</b> |                      |                                  |             |              |
|------------------|----------------------|----------------------------------|-------------|--------------|
| 0613 554<br>07A  | ICT/CU/CS/CC/01/6/MA | Basic Electronic Skills          | 180         | 18.0         |
| 0613 554<br>08A  | ICT/CU/CS/CC/02/6/MA | Fundamentals of Programming      | 200         | 20.0         |
| 0417 441<br>02A  | ICT/CU/CS/BC/02/6/MA | Work Ethics and Practices        | 40          | 4.0          |
|                  |                      | <b>SUB TOTAL</b>                 | <b>420</b>  | <b>42.0</b>  |
| <b>MODULE V</b>  |                      |                                  |             |              |
| 0413 441<br>03A  | ICT/CU/CS/BC/03/6/MA | Entrepreneurial Skills           | 40          | 4.0          |
| 0613 554<br>09A  | ICT/CU/CS/CR/07/6/MA | Algorithms and Data Structures   | 190         | 19.0         |
| 0613 554<br>10A  | ICT/CU/CS/CC/03/6/MA | Mathematics for Computer Science | 170         | 17.0         |
|                  |                      | <b>SUB TOTAL</b>                 | <b>400</b>  | <b>40.0</b>  |
| <b>MODULE VI</b> |                      |                                  |             |              |
| 0031 441<br>01A  | ICT/CU/CS/BC/01/6/MA | Communication Skills             | 40          | 4.0          |
| 0613 554<br>11A  | ICT/CU/CS/CR/08/6/MA | Information Systems Development  | 200         | 20.0         |
| 0613 554<br>12A  | ICT/CU/CS/CC/04/6/MA | Artificial Intelligence Concepts | 180         | 18.0         |
|                  |                      | <b>SUB TOTAL</b>                 | <b>420</b>  | <b>42.0</b>  |
|                  | ICT/CU/CS/CR/09/6/MA | <b>INDUSTRY TRAINING</b>         | <b>480</b>  | <b>48.0</b>  |
|                  |                      | <b>GRAND TOTAL</b>               | <b>2980</b> | <b>298.0</b> |

## **Entry Requirements**

An individual entering this course should have any of the following minimum requirements:

- a) Kenya Certificate of Secondary Education (KCSE) C- (minus)
- b) Equivalent qualifications as determined by relevant regulatory body

## **Trainer Qualification**

Qualifications of a trainer for this course include:

- a) Possession of at least Computer Science level 7 or in related trade area;
- b) License by TVETA; and
- c) License by regulatory body (where applicable)

## **Industry Training**

An individual enrolled in this course will be required to undergo Industry training for a minimum period of 480 hours in ICT sector. The industrial training may be taken after completion of all units for those pursuing the full qualification or be distributed equally in each unit for those pursuing part qualification. In the case of dual training model, industrial training shall be as guided by the dual training policy.

## **Assessment**

The course shall be assessed formatively and summative:

- a) During formative assessment all performance criteria shall be assessed based on performance criteria weighting.
- b) Number of formative assessments shall minimally be equal to the number of elements in a unit of competency.
- c) During summative assessment basic and common units may be integrated in the core units or assessed as discrete units.
- d) Theoretical and practical weighting for each unit of learning shall be 40-60 for all units.
- e) Formative and summative assessments shall be weighted at 60% and 40% respectively in the overall unit of learning score

For a candidate to be declared competent in a unit of competency, the candidate must

meet the following conditions:

- i) Obtained at least 40% in theory assessment in formative and summative assessments.
  - ii) Obtained at least 60% in practical assessment in formative and summative assessment where applicable.
  - iii) Obtained at least 50% in the weighted results between formative assessment and summative assessment where the former constitutes 60% and the latter 40% of the overall score.
- f) Assessment performance rating for each unit of competency shall be as follows:

| MARKS        | COMPETENCE RATING                     |
|--------------|---------------------------------------|
| 80 -100      | Attained Mastery                      |
| 65 - 79      | Proficient                            |
| 50 - 64      | Competent                             |
| 49 and below | Not Yet Competent                     |
| Y            | Assessment Malpractice/irregularities |

- g) Assessment for Recognition of Prior Learning (RPL) may lead to award of part and/or full qualification.

## Certification

A candidate will be issued with a Certificate of Competency upon demonstration of competence in a core Unit of Competency. To be issued with Kenya National TVET Certificate in Computer Science level 6, the candidate must demonstrate competence in all the Units of Competency as given in the qualification pack. Statement of Attainment certificate may be awarded upon demonstration of competence in certifiable element within a unit.

These certificates will be issued by TVET CDACC

## **MODULE I**

## COMPUTER ORGANISATION AND ARCHITECTURE

**ISCED UNIT CODE:** 0613 554 01A

**UNIT CODE:** ICT/CU/CS/CR/01/6/MA

### Relationship to Occupational Standards

This unit addresses the unit of competency: Analyse Computer Organization and Architecture

**Duration of Unit:** 180 hours

### Unit description

This unit covers the competencies required to understand computer organisation and architecture. It involves understanding principles of computer organisation and design, understanding central processing unit functions, understanding computer memory organization, understanding input-output functions and understanding computer arithmetic and logic.

### Summary of Learning Outcomes

| Learning Outcomes                                 | Durations (Hours) |
|---------------------------------------------------|-------------------|
| 1. Principles of computer organization and design | 30                |
| 2. Central processing unit functions              | 35                |
| 3. Computer memory functions                      | 35                |
| 4. Input-output functions                         | 30                |
| 5. Computer arithmetic and logic                  | 50                |
| <b>TOTAL</b>                                      | <b>180</b>        |

### Learning Outcomes, Content and Suggested Assessment Methods

| Learning Outcome          | Content                                                                                     | Suggested Assessment Methods                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 1. Principles of computer | 1.1 ICT concepts<br>1.2 Principles of computer organisation<br>1.3 Computer architecture is | <ul style="list-style-type: none"><li>• Practical tests</li><li>• Observation</li><li>• Oral tests</li></ul> |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                           |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| organisation and design              | 1.4 Structure and function of <i>computer hardware components</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Written tests</li> </ul>                                                                         |
| 2. Central processing unit functions | 2.1 Peripheral devices<br>2.2 The Central Processing Unit<br>2.3 CPU architecture<br>2.4 Registers<br>2.5 Instruction representation and execution<br>2.6 CPU specifications<br>2.7 CPU specifications are verification of computer system                                                                                                                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Observation</li> <li>• Oral tests</li> <li>• Written tests</li> </ul> |
| 3. Computer memory organization      | 3.1 Computer Memory Organization <ul style="list-style-type: none"> <li>3.1.1 Functions</li> <li>3.1.2 Categories of internal memory</li> <li>3.1.3 Standard memory specification factors</li> </ul> 3.2 Storage technologies <ul style="list-style-type: none"> <li>3.2.1 Solid state storage devices</li> <li>3.2.2 Optical storage devices</li> <li>3.2.3 Magnetic storage devices</li> </ul> 3.3 Cache and Virtual memory <ul style="list-style-type: none"> <li>3.3.1 Definitions</li> <li>3.3.2 Operations of cache and virtual memory</li> </ul> 3.4 Prescription of memory specifications as per user needs<br>Verification of memory specifications for a given computer | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Observation</li> <li>• Oral tests</li> <li>• Written tests</li> </ul> |
| 4. Input-output organization         | 4.1 Peripheral devices<br>4.2 Standard I/O devices specification factors                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Observation</li> </ul>                                                |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                           |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | 4.3 Role of Bus interface in I/O<br>4.4 Modes of data transfer<br>2.4.1 Programmed I/O<br>2.4.2 Interrupt initiated I/O<br>2.4.3 Direct memory access (DMA)<br>4.5 I/O devices' specifications as per user needs<br>4.6 Verification of computer I/O devices' specifications<br>4.7 Input-output processing                                                                                                                                         | <ul style="list-style-type: none"> <li>• Oral tests</li> <li>• Written tests</li> </ul>                                                   |
| 5. Computer arithmetic and logic | 5.1 Number systems<br>5.1.1 Types<br>5.1.2 Operations<br>5.1.3 Conversion<br>5.2 IEEE-based Integer and Floating point representations<br>5.3 Integer and Floating-point arithmetic<br>5.3.1 Addition<br>5.3.2 Subtraction<br>5.3.3 Multiplication<br>5.4 Logic operators<br>5.4.1 OR<br>5.4.2 AND<br>5.4.3 NAND<br>5.4.4 NOR<br>5.4.5 NOT<br>5.5 Logic operations<br>5.5.1 Addition<br>5.5.2 Multiplication<br>5.5.3 Subtraction<br>5.5.4 Division | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Observation</li> <li>• Oral tests</li> <li>• Written tests</li> </ul> |



|  |                                                                                                    |  |
|--|----------------------------------------------------------------------------------------------------|--|
|  | 5.6 Representing logic operations<br>5.6.1 Truth table<br>5.6.2 Karnaugh maps<br>5.6.3 Logic gates |  |
|--|----------------------------------------------------------------------------------------------------|--|

### Suggested Methods of Instruction

- Presentations and practical demonstrations by trainer;
- Guided learner activities and research to develop underpinning knowledge;
- Supervised activities and projects in a workshop
- Simulation
- Visiting lecturer/specialist from the ICT sector;
- Industrial visits.

### Recommended Resources for 25 trainees

| S/No. | Category/Item             | Description/Specifications                                                                        | Quantity | Recommended Ratio (Trainee: Item) |
|-------|---------------------------|---------------------------------------------------------------------------------------------------|----------|-----------------------------------|
| 1     | Desktop Computers         | Minimum: Intel i5 or Ryzen 5, 8GB RAM, 256GB SSD, USB ports, HDMI/VGA, Windows/Linux OS installed | 13       | 2:1                               |
| 2     | Projector                 | HD resolution, HDMI/VGA input.                                                                    | 1        | 25:1                              |
| 3     | Whiteboard & Markers      | Whiteboard (min 4x6 feet), assorted color markers                                                 | 1 set    | 25:1                              |
| 4     | Computer Hardware Toolkit | Screwdrivers, anti-static wristbands, pliers, cable testers, etc.                                 | 5 sets   | 5:1                               |

|    |                            |                                                                                        |                  |                                            |
|----|----------------------------|----------------------------------------------------------------------------------------|------------------|--------------------------------------------|
| 5  | Sample Hardware Components | CPU (old and new), RAM (DDR3/DDR4), SSD/HDD, Motherboard, GPU, PSU, I/O cards          | 5 sets           | 5:1                                        |
| 6  | Peripheral Devices         | Keyboards, mice, scanners, webcams, printers (shared), microphones, external drives    | 25 sets (shared) | 1:1 (essential); shared for optional items |
| 7  | Networking Devices         | Switches, routers, Ethernet cables, RJ45 crimping tools                                | 2 sets           | Shared for demonstrations                  |
| 8  | Internet Access            | Broadband/Wi-Fi for software downloads, online tools                                   | 1                | 25:1                                       |
| 9  | Simulation Software        | Logic simulator (e.g. Logisim), CPU simulators, memory management simulators           | Site License     | 25:1 (individual access)                   |
| 10 | Number System Charts       | Binary, Octal, Decimal, Hexadecimal conversion tables                                  | 5 sets           | 5:1                                        |
| 11 | Printed Handouts/Manuals   | Module notes, exercises, CPU cycle diagrams, logic gate explanations, memory hierarchy | 25               | 1:1                                        |
| 12 | Storage Devices            | Flash drives for each trainee                                                          | 25               | 1:1                                        |
| 13 | Karnaugh Map Templates     | Printed sheets or laminated reusable versions                                          | 10 sets          | 2.5:1                                      |
| 14 | Logic Gate Kits            | Breadboards, logic ICs (AND, OR, NOT, NAND, NOR, XOR), LEDs, resistors, power supply   | 10 sets          | 2.5:1                                      |

|    |                            |                                                                                                       |              |        |
|----|----------------------------|-------------------------------------------------------------------------------------------------------|--------------|--------|
| 15 | Digital Multimeters        | For circuit testing and diagnostics                                                                   | 5            | 5:1    |
| 16 | Instructor Computer/Laptop | Higher spec than trainee systems, used for demos, presentations                                       | 1            | 25:1   |
| 17 | Software Licenses          | OS, simulation tools, productivity software (MS Office or LibreOffice), compilers (GCC, Python, etc.) | Site License | 25:1   |
| 18 | Backup Storage Device      | External HDD/SSD (1TB or more) for central data backup                                                | 1            | Shared |
| 19 | Printer                    | Laser or inkjet printer with scanning functionality                                                   | 1            | Shared |
| 20 | Stationery Supplies        | Pens, notebooks, sticky notes, highlighters, rulers, etc.                                             | 25 sets      | 1:1    |

## OPERATING SYSTEMS CONFIGURATION

**ISCED UNIT CODE:** 0613 554 02A

**UNIT CODE:** ICT/CU/CS/CR/02/6/MA

### Relationship to Occupational Standards

This unit addresses the unit of competency: Configure Operating Systems

**Duration of Unit:** 240 hours

### Unit Description:

This unit covers the competencies required to understand operating systems. It involves understanding fundamentals of operating systems, applying computer application softwares to solving tasks, understanding process management, understanding memory management, understanding input-output management and understanding file management.

### Summary of Learning Outcomes:

| Learning Outcomes                                 | Durations (Hours) |
|---------------------------------------------------|-------------------|
| 1. Computer application software to solving tasks | 70                |
| 2. Process management                             | 25                |
| 3. Memory management                              | 25                |
| 4. Input and output management                    | 40                |
| 5. File management and local policy settings      | 40                |
| <b>TOTAL</b>                                      | <b>240</b>        |

### Learning Outcomes, Content and Suggested Assessment Methods

| Learning Outcome                     | Content                                                                                   | Suggested Assessment Methods                                                                                                             |
|--------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Fundamentals of operating systems | 1.1 Computer software<br>1.1.1 Definition<br>1.1.2 Classification<br>1.2 Operating system | <ul style="list-style-type: none"><li>• Practical exercises</li><li>• Oral tests</li><li>• Written tests</li><li>• Observation</li></ul> |

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                             |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <p>1.2.1 Definition</p> <p>1.2.2 Concepts</p> <p>1.2.3 Functions of operating system are identified.</p> <p>1.3 Operating system structures</p> <p>1.3.1 Monolithic</p> <p>1.3.2 Layered</p> <p>1.3.3 Virtual</p> <p>1.3.4 Client-server model</p> <p>1.4 Types of operating systems</p> <p>1.5 Requirements for Windows OS installation</p> <p>1.6 Demonstration of Windows installation</p> <p>1.6.1 Specify hardware requirements</p> <p>1.6.2 Back up data in target machine</p> <p>1.6.3 Partition creation and/or formatting</p> <p>1.6.4 Installation as per vendor instructions</p> <p>1.6.5 Testing installation</p> |                                                                                                                                             |
| 2. Computer application software for solving tasks | <p>2.1 Word processing</p> <p>2.1.1 Functions and concepts of word processing.</p> <p>2.1.2 Documents and tables creation and manipulations</p> <p>2.1.3 Mail merging</p> <p>2.1.4 Word processing utilities</p> <p>2.1.5 Printing</p> <p>2.2 Presentation Packages;</p>                                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>• Oral questioning</li> <li>• Project</li> <li>• Practical tests</li> <li>• Written tests</li> </ul> |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>2.2.1 Types of presentation Packages</p> <p>2.2.2 Creating, formulating, running, editing, printing and presenting slides and handouts</p> <p>2.3 Spread sheets</p> <p>2.3.1 Meaning, formulae, function and charts, uses and layout</p> <p>2.3.2 Data formulation, manipulation and application to cells</p> <p>2.3.3 Printing</p> <p>2.4 Database design and manipulation</p> <p>2.4.1 Database design, data manipulation, sorting, indexing, storage retrieval and security</p> <p>2.5 Data manipulation, storage and retrieval</p> <p>2.6 Office Internet and Electronic mail;</p> <p>2.6.1 Office internet Connectivity</p> <p>2.6.2 Internet Browsing</p> <p>2.6.3 Electronic mail</p> |                                                                                                                                               |
| 3. Process management | <p>3.1 Process management</p> <p>3.1.1 Definitions: Process, Thread, Process Control Block</p> <p>3.1.2 Functions of the Process Manager</p> <p>3.2 Computer Resources</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>• Practical exercises</li> <li>• Oral tests</li> <li>• Written tests</li> <li>• Observation</li> </ul> |

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|                                | <p>3.3 Process states and their transition</p> <p>3.3.1 States: Ready, Waiting, Complete, Running</p> <p>3.3.2 Transitions: Dispatch, Suspend, Exit, Resume</p> <p>3.4 Process scheduling</p> <p>3.4.1 Features of scheduling algorithms</p> <p>3.4.2 Types of schedulers</p> <p>3.4.3 Scheduling algorithms</p> <p>3.5 Demonstration of Task Manager</p> <p>3.5.1 Observing CPU queue</p> <p>3.5.2 Stopping CPU intensive processes.</p> <p>3.6 Performance monitor tools in process management</p> |                                                                                                                                               |
| 4. Memory management           | <p>4.1 Memory Management</p> <p>4.1.1 Definition</p> <p>4.1.2 Objectives of Memory management</p> <p>4.1.3 Components of the Memory Management unit</p> <p>4.2 Memory management techniques</p> <p>4.2.1 Partitioning</p> <p>4.2.2 Virtual memory:</p> <p>4.2.3 Paging, Segmentation</p> <p>4.3 Demonstration of virtual memory settings – Increasing the Windows page file size</p>                                                                                                                 | <ul style="list-style-type: none"> <li>• Practical exercises</li> <li>• Oral tests</li> <li>• Written tests</li> <li>• Observation</li> </ul> |
| 5. Input and output management | <p>5.1 Input - output management</p> <p>5.1.1 Definition</p>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>• Practical exercises</li> <li>• Oral tests</li> <li>• Written tests</li> </ul>                        |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                 |
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|  | <p>5.1.2 Objectives of I/O management</p> <p>5.1.3 I/O hardware</p> <p>5.1.4 I/O software</p> <p>5.1.5 Polling Vs Interrupt drive I/O</p> <p>5.2 Disk operations</p> <p>5.2.1 Access time factors</p> <p>5.2.2 Techniques for resolving slow disk I/O</p> <p>5.3 Computer clock system</p> <p>5.3.1 Virtual Input Output</p> <p>5.3.2 Definition of Virtual I/O</p> <p>5.3.3 Types of virtual I/O: Buffering, Spooling, Caching</p> <p>5.4 Disk selection criteria</p> <p>5.4.1 Size</p> <p>5.4.2 Speed</p> <p>5.5 Demonstration of disk storage management operations</p> <p>5.5.1 Formatting volume</p> <p>5.5.2 Partitioning volume</p> <p>5.5.3 Shrinking volume</p> <p>5.5.4 Extending volume</p> <p>5.5.5 Optimising and defragmenting disk</p> <p>5.5.6 Changing drive security permissions</p> <p>5.5.7 Backing up</p> <p>5.5.8 Copying data to optical disks</p> <p>5.6 Handling removable media</p> | <ul style="list-style-type: none"> <li>• Observation</li> </ul> |
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|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                               |
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|                    | 5.6.1 Demonstration of device management operations using Windows Device Manager<br>5.6.2 Verifying installed drivers<br>5.6.3 Resolving driver conflicts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                               |
| 6. File management | 6.1 File management<br>6.1.1 Definition<br>6.1.2 Objectives of file manager<br>6.1.3 File naming concepts<br>6.2 File access methods<br>6.2.1 Sequential access<br>6.2.2 Direct/Random access<br>6.2.3 Indexed sequential access<br>6.3 File allocation techniques<br>6.3.1 Contiguous<br>6.3.2 File Allocation<br>6.3.3 Indexed<br>6.4 File protection and security<br>6.4.1 Importance<br>6.4.2 Access control<br>6.4.3 Audit trail<br>6.5 File and directory operations<br>6.5.1 Creating folders and files<br>6.5.2 Renaming folders and files<br>6.5.3 Deleting folders and files<br>6.5.4 Copying and Moving folders and files<br>6.5.5 Setting file attributes<br>6.6 Local security policy settings<br>6.6.1 Password policy<br>6.6.2 Account lockout policy | <ul style="list-style-type: none"> <li>• Practical exercises</li> <li>• Oral tests</li> <li>• Written tests</li> <li>• Observation</li> </ul> |

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|--|------------------------|--|
|  | 6.6.3 Audit policy     |  |
|  | 6.6.4 Security options |  |

### Suggested Methods of Instruction

- Presentations and practical demonstrations by trainer;
- Guided learner activities and research to develop underpinning knowledge;
- Supervised activities and projects in a workshop;
- Visiting lecturer/trainer from the ICT sector;
- Industrial visits.

### Recommended Resources for 25 trainees

| S/No. | Category/Item              | Description/Specifications                         | Quantity                        | Recommended Ratio (Trainee: Item) |
|-------|----------------------------|----------------------------------------------------|---------------------------------|-----------------------------------|
| 1     | <b>Learning Materials</b>  | OS & Application Software textbook / printed notes | 25 copies                       | 1:1                               |
| 2     |                            | Training manual                                    | 25 copies                       | 1:1                               |
| 3     |                            | Worksheets, quizzes, and assignments               | 25 sets                         | 1:1                               |
| 4     | <b>Learning Facilities</b> | Training room with seating and power sockets       | For 25 trainees + 1 instructors | N/A                               |
| 5     |                            | Whiteboard/Smartboard with markers or projector    | 1                               | 1 per class                       |
| 6     | <b>Infrastructure</b>      | Reliable electricity and backup (UPS or generator) | 1 setup                         | Shared                            |
| 7     |                            | Networked computer lab with LAN/internet access    | 1 lab                           | Shared                            |
| 8     |                            | Server system for demo                             | 1                               | 1 per lab                         |

|    |                              |                                                       |                          |             |
|----|------------------------------|-------------------------------------------------------|--------------------------|-------------|
| 9  | <b>Tools &amp; Equipment</b> | Desktop/Laptop computers                              | 25                       | 1:1         |
| 10 |                              | Printer (LaserJet or Inkjet)                          | 1–2                      | 10–15:1     |
| 11 |                              | Projector and screen                                  | 1                        | 1 per class |
| 12 |                              | External storage (USBs or external HDDs)              | 10                       | 2–3:1       |
| 13 |                              | Optical drives (USB DVD writer)                       | 5                        | 5:1         |
| 14 |                              | Network Switch + Cables                               | 1 setup                  | Shared      |
| 15 |                              | Internet Connectivity (Wi-Fi or wired LAN)            | 1 setup                  | Shared      |
| 16 |                              | Office application software (MS Office / LibreOffice) | 25 licenses              | 1:1         |
| 17 |                              | Virtualization software (VMware, VirtualBox)          | 25 installations         | 1:1         |
| 18 |                              | Operating Systems (Windows/Linux ISO images)          | 25 copies                | 1:1         |
| 19 |                              | Antivirus software                                    | 25 licenses              | 1:1         |
| 20 | <b>Consumable Materials</b>  | Writing pads, pens                                    | 25 sets                  | 1:1         |
| 21 |                              | Printer ink and paper                                 | As required              | Shared      |
| 22 |                              | Backup media (DVDs, USB sticks)                       | 25                       | 1:1         |
| 23 | <b>Instructor Materials</b>  | Instructor laptop with teaching software              | 1                        | 1:1         |
| 24 |                              | Lesson plans, slides, and teaching aids               | 1 set                    | 1:1         |
| 25 | <b>Support Tools</b>         | Windows Device Manager & Task Manager tools           | Installed on each system | 1:1         |
| 26 |                              | Performance Monitor, Disk Management tools            | Installed on each system | 1:1         |

## **MODULE II**

## NETWORKING AND DISTRIBUTED SYSTEMS

**ISCED UNIT CODE:** 0613 554 03A

**UNIT CODE:** ICT/CU/CS/CR/07/6/MA

### Relationship to Occupational Standards

This unit addresses the unit of competency: Create networks and Distributed Systems

**Duration of Unit:** 220 hours

### Unit description:

This unit specifies the competencies required to understanding networking and distributed systems concept. It involves understanding networking and distributed systems, distributed system architectures, distributed processing and file management, setting up a network in a distributed environment understanding data communication standards and IP addressing and troubleshooting a network.

### Summary of Learning Outcomes

| Learning Outcomes                                 | Durations (Hours) |
|---------------------------------------------------|-------------------|
| 1. Networking and distributed systems concepts    | 25                |
| 2. Distributed systems architectures              | 30                |
| 3. Distributed processing and file management     | 30                |
| 4. Network in a distributed environment           | 55                |
| 5. Data Communication Standards and IP addressing | 30                |
| 6. Network troubleshoot                           | 50                |
| <b>TOTAL</b>                                      | <b>220</b>        |

### Learning Outcomes, Content and Suggested Assessment Methods

| Learning Outcome                      | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Suggested Assessment Methods                                                                                                              |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Networking and distributed systems | 1.1 Fundamentals of networking<br>1.1.1 Definition of network<br>1.1.2 Definition of network terminologies<br>1.1.3 Identified network components<br>1.1.4 Application and benefits of networking<br>1.2 Types of networks<br>1.2.1 LAN<br>1.2.2 MAN<br>1.2.3 WAN<br>1.2.4 PAN<br>1.3 Network topologies<br>1.3.1 Star<br>1.3.2 Ring<br>1.3.3 Mesh<br>1.3.4 Bus<br>1.4 Transmission media<br>1.4.1 Wired media<br>1.4.2 Wireless media<br>1.5 Distributed system<br>1.5.1 Definition<br>1.5.2 Application<br>1.6 Types of distributed systems<br>1.6.1 Computing<br>1.6.2 Information<br>1.6.3 Pervasive<br>1.6.4 Client server<br>1.6.5 Peer to peer | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Observation</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                           |
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|                                     | <p>1.7 Distributed systems models</p> <p>1.7.1 Architectural</p> <p>1.7.2 Interaction</p> <p>1.7.3 Fault</p> <p>1.8 Specifying network requirements for a site</p> <p>1.8.1 Type of network</p> <p>1.8.2 Type of topology</p> <p>1.8.3 Devices</p> <p>1.9 Network security</p> <p>1.9.1 Definition</p> <p>1.9.2 Types of network attacks</p> <p>1.9.2.1 Active</p> <p>1.9.2.2 Passive</p> <p>1.10 Components of network security</p> <p>1.10.1 Network access control</p> <p>1.10.2 Firewall</p> <p>1.10.3 Intrusion prevention</p> <p>1.10.4 Security information and event management</p> <p>1.11 Wireless security</p> |                                                                                                                                           |
| 2 Distributed systems architectures | <p>2.1 Distributed architecture</p> <p>2.1.1 Definition</p> <p>2.1.2 Application</p> <p>2.2 Architecture styles</p> <p>2.2.1 Layered Architecture</p> <p>2.2.2 Object Based Architecture</p> <p>2.2.3 Data-centred Architecture</p> <p>2.3 Types of distributed system architectures</p> <p>2.3.1 Centralized</p> <p>2.3.2 Decentralized</p>                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Observation</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |

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|                                                  | <p>2.3.3 Hybrid</p> <p>2.4 Specifying distributed system architecture requirements for a simulated site</p> <p>2.4.1 Architecture style</p> <p>2.4.2 Type of distributed system architectures</p>                                                                                                  |                                                                                                                                           |
| 3 Distributed processing and file management     | <p>3.1 Types of distributed processing</p> <p>3.1.1 Distributed processing</p> <p>3.1.2 Parallel processing</p> <p>3.2 Types of file systems</p> <p>3.3 File sharing and accessing methods</p> <p>3.3.1 Remote access</p> <p>3.3.2 Data caching</p> <p>3.4 Distributed file sharing and access</p> | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Observation</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |
| 4 Network Set up in a distributed environment    | <p>4.1 Tools, materials and devices</p> <p>4.2 Connection and configuration of network devices</p> <p>4.3 Installation and configuration of network software</p> <p>4.4 Testing the network</p>                                                                                                    | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Observation</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |
| 5 Data Communication standards and IP addressing | <p>5.1 OSI model</p> <p>5.1.1 Definition</p> <p>5.1.2 Functions of different OSI model layers</p> <p>5.1.3 OSI layer Protocols are illustrated</p> <p>5.2 Data communication components</p> <p>5.2.1 Message</p> <p>5.2.2 Sender</p> <p>5.2.3 Receiver</p>                                         |                                                                                                                                           |



|                           |                                                                                                                                                                                                                            |                                                                                                                                           |
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|                           | 5.2.4 Medium<br>5.2.5 Protocol<br>5.3 Network IP Address classes<br>5.3.1 Class A, B, C<br>5.3.2 Public and Private IP Address<br>5.3.3 Automatic Private IP Address                                                       |                                                                                                                                           |
| 6 Network Troubleshooting | 6.1 Troubleshooting techniques<br>6.1.1 Definition<br>6.1.2 Techniques<br>6.1.3 Procedures<br>6.2 Troubleshooting tools<br>6.2.1 Ping<br>6.2.2 Tracert/traceroute<br>6.2.3 Nslookup<br>6.2.4 Netstat<br>6.2.5 Pathping/mtr | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Observation</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |

### Suggested Methods of Instruction

- Presentations and practical demonstrations by trainer;
- Guided learner activities and research to develop underpinning knowledge;
- Supervised activities and projects in a site;
- Visiting lecturer/trainer from the ICT sector;
- Industrial visits.

### Recommended Resources for 25 trainees

| S/No. | Category/Item | Description/Specifications | Quantity | Recommended Ratio (Trainee: Item) |
|-------|---------------|----------------------------|----------|-----------------------------------|
|       |               |                            |          |                                   |

|    |                              |                                                                                  |           |                          |
|----|------------------------------|----------------------------------------------------------------------------------|-----------|--------------------------|
| 1  | <b>Learning Materials</b>    | Networking and Distributed Systems textbooks or manuals                          | 25 copies | 1:1                      |
| 2  |                              | OSI Model diagrams, IP classes charts, and network topology charts               | 25 sets   | 1:1                      |
| 3  |                              | Printed lab exercises and simulations                                            | 25 sets   | 1:1                      |
| 4  |                              | Practical guides for configuring routers, switches, and distributed file systems | 25 sets   | 1:1                      |
| 5  | <b>Learning Facilities</b>   | Networked computer lab room with furniture, ventilation, power sockets           | 1 lab     | Shared                   |
| 6  |                              | Whiteboard/Smartboard or Projector + Screen                                      | 1         | 1 per class              |
| 7  |                              | Instructor station with demo setup and internet access                           | 1         | 1:1                      |
| 8  | <b>Infrastructure</b>        | Reliable electricity + Backup (UPS or Generator)                                 | 1 setup   | Shared                   |
| 9  |                              | Local Area Network (LAN) setup with internet access                              | 1 setup   | Shared                   |
| 10 |                              | Internet access and router with DHCP and static IP configuration capability      | 1 setup   | Shared                   |
| 11 | <b>Tools &amp; Equipment</b> | Computers with NICs (Ethernet, Windows/Linux)                                    | 25        | 1:1                      |
| 12 |                              | Network cables (CAT6 or CAT5e)                                                   | 30+       | 1 per PC + extras        |
| 13 |                              | Ethernet switches (8/16/24 port unmanaged or managed)                            | 2–3       | 8–12 trainees per switch |
| 14 |                              | Wireless routers (for Wi-Fi and access point configuration)                      | 2         | 12–15:1                  |

|    |                                |                                                                                      |                            |        |
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| 15 |                                | Patch panels and cable testers                                                       | 1–2                        | Shared |
| 16 |                                | Crimping tools + RJ45 connectors                                                     | 5 sets                     | 5:1    |
| 17 |                                | PCs/laptops with virtualization software (VMware/VirtualBox) for distributed systems | 25                         | 1:1    |
| 18 |                                | Pre-installed network simulation software (e.g., Cisco Packet Tracer, GNS3)          | 25 installations           | 1:1    |
| 19 |                                | Operating system ISOs (Windows Server, Ubuntu, etc.)                                 | 25 copies                  | 1:1    |
| 20 |                                | IP address planning sheets and subnetting charts                                     | 25                         | 1:1    |
| 21 | <b>Consumable Materials</b>    | RJ45 connectors, electrical tape, velcro straps                                      | As needed                  | Shared |
| 22 |                                | Writing pads, pens, markers                                                          | 25 sets                    | 1:1    |
| 23 |                                | Printing paper and ink                                                               | As needed                  | Shared |
| 24 | <b>Software &amp; Security</b> | Firewall and antivirus software for training labs                                    | 25 licenses                | 1:1    |
| 25 |                                | Packet sniffing tools (Wireshark)                                                    | 25 installations           | 1:1    |
| 26 |                                | Network troubleshooting tools (Ping, Traceroute, Nslookup, Netstat, Pathping)        | Preinstalled               | 1:1    |
| 27 | <b>Demonstration Kits</b>      | Router and switch demo kits                                                          | 2–3                        | 8–12:1 |
| 28 |                                | Distributed file system demonstration setup (e.g., shared folders, Samba, NFS)       | Configured on lab machines | 1:1    |
| 29 |                                | Network security demo tools (SIEM dashboard, simulated firewall, IDS/IPS)            | 1 virtual lab              | Shared |

|    |                             |                                                                          |       |     |
|----|-----------------------------|--------------------------------------------------------------------------|-------|-----|
| 30 | <b>Instructor Materials</b> | Instructor's PC/laptop, preloaded with network scenarios and simulations | 1     | 1:1 |
| 31 |                             | Lesson plans, assessment sheets, observation checklists                  | 1 set | 1:1 |

## GRAPHIC DESIGN

**ISCED UNIT CODE:** 0613 554 04A

**UNIT CODE:** ICT/CU/CS/CR/11/6/MA

### Relationship to Occupational Standards

This unit addresses the unit of competency: Design Graphics

**Duration of Unit:** 190 hours

### Unit description:

This unit covers the competencies required to understand Graphic Design. It involves understanding fundamentals of graphic design, understanding elements and principles of graphic design, applying typography techniques, creating and editing of images, performing layout design and printing the design.

### Summary of Learning Outcomes

| Learning Outcomes                            | Durations (Hours) |
|----------------------------------------------|-------------------|
| 1. Fundamentals of graphic design            | 20                |
| 2. Elements and principles of graphic design | 35                |
| 3. Typography techniques                     | 25                |
| 4. Image Creation and editing                | 45                |
| 5. Layout design                             | 45                |
| 6. Print design                              | 20                |
| <b>TOTAL</b>                                 | <b>190</b>        |

## Learning Outcomes, Content and Suggested Assessment Methods

| Learning Outcome                             | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Suggested Assessment Methods                                                                                                              |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Graphic design fundamentals               | 1.1 Graphic Design fundamentals <ul style="list-style-type: none"> <li>1.1.1 Definition</li> <li>1.1.2 Types of elements</li> <li>1.1.3 Principles</li> <li>1.1.4 Application areas</li> </ul> 1.2 Graphic design equipment <ul style="list-style-type: none"> <li>1.2.1 Computer</li> <li>1.2.2 Scanner</li> <li>1.2.3 Printer</li> <li>1.2.4 Camera</li> <li>1.2.5 Digital Tablet</li> </ul> 1.3 Specified requirements as per user requirements<br>1.4 Application areas of graphic design | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Observation</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |
| 2. Elements and principles of graphic design | 2.1 Demonstration of elements <ul style="list-style-type: none"> <li>2.1.1 Colour</li> <li>2.1.2 Line</li> <li>2.1.3 Space</li> <li>2.1.4 Shape</li> <li>2.1.5 Texture</li> <li>2.1.6 Value</li> </ul> 2.2 Principles of graphic design <ul style="list-style-type: none"> <li>2.2.1 Balance</li> <li>2.2.2 Contrast</li> <li>2.2.3 Emphasis</li> <li>2.2.4 Harmony</li> <li>2.2.5 Pattern</li> <li>2.2.6 Proportion</li> </ul>                                                               | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Observation</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |

|                               |                                                                                                                                                                                                                                         |                                                                                                                                           |
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|                               | 2.2.7 Unity<br>2.3 Selected appropriate elements for graphic design project                                                                                                                                                             |                                                                                                                                           |
| 3. Typography techniques      | 3.1 Typography techniques<br>3.1.1 Definition<br>3.1.2 Types of techniques<br>3.2 Measurements and standards<br>3.3 Selecting an appropriate typography technique for graphic design project                                            | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Observation</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |
| 4. Image Creation and editing | 4.1 Identification of graphic design and photography Software and tools<br>4.2 Image file types<br>3.3.1 Raster<br>3.3.2 Vector<br>4.3 Creation of letter forms, lines of type and body copy<br>4.4 Creation and manipulation of images | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Observation</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |
| 5. Design layout              | 5.1 Proportion on layout design<br>5.2 Creation of unified systems out of dissimilar elements<br>5.3 Creation of dynamic layouts using typographic tools<br>5.4 Creation of Type and image project<br>5.5 Multi-page layout planning    | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Observation</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |
| 6. Print design               | 6.1 Printing tools and Equipment<br>6.2 Types of printing<br>6.3 Paper classification<br>6.3.1 Types<br>6.3.2 Size                                                                                                                      | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Observation</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |

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|  | 6.3.3 Weight<br>6.4 Selection of printing chemicals<br>6.5 Demonstration of actual design printing |  |
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### Suggested Methods of Instruction

- Presentations and practical demonstrations by trainer;
- Guided learner activities and research to develop underpinning knowledge;
- Supervised practical activities and projects
- Visiting lecturer/trainer from the ICT sector;
- Industrial visits.

### Recommended Resources for 25 trainees

| S/No. | Category                   | Item Description / Specifications                                                   | Quantity | Recommended Ratio (Trainee: Item) |
|-------|----------------------------|-------------------------------------------------------------------------------------|----------|-----------------------------------|
| 1     | <b>Learning Materials</b>  | Graphic Design Textbooks, Typography Guides, Layout Design Manuals, Printing Guides | 25       | 1:1                               |
| 2     |                            | Access to Online Learning Platforms                                                 | 25       | 1:1                               |
| 3     | <b>Learning Facilities</b> | Training Room with 25 Workstations, AC, Lighting, Whiteboard, Projector             | 1 Room   | -                                 |
| 4     |                            | Multimedia Projector                                                                | 1        | 1:25                              |
| 5     |                            | Ergonomic Chairs and Desks                                                          | 25 Sets  | 1:1                               |
| 6     | <b>Infrastructure</b>      | Stable High-Speed Internet Connection (Minimum 100 Mbps)                            | 1 Setup  | -                                 |
| 7     |                            | Backup Power Supply / Generator                                                     | 1 Unit   | -                                 |



|    |                                 |                                                                               |             |        |
|----|---------------------------------|-------------------------------------------------------------------------------|-------------|--------|
| 8  | <b>Tools &amp; Equipment</b>    | Desktop Computers / Laptops                                                   | 25          | 1:1    |
| 9  |                                 | Flatbed Scanners (A4/A3)                                                      | 5           | 1:5    |
| 10 |                                 | Printers (Inkjet/Laser, A3 Capability Preferred)                              | 3           | 1:8    |
| 11 |                                 | DSLR Digital Cameras                                                          | 5           | 1:5    |
| 12 |                                 | Digital Drawing Tablets (Wacom or equivalent, A5 or larger)                   | 10          | 1:2.5  |
| 13 | <b>Software</b>                 | Adobe Creative Cloud (Photoshop, Illustrator, InDesign, etc.) Licenses        | 25          | 1:1    |
| 14 |                                 | Typography Software/Font Libraries (e.g., Adobe Fonts, Google Fonts)          | 1 Access    | Shared |
| 15 |                                 | Page Layout Software (InDesign, Affinity Publisher)                           | 25          | 1:1    |
| 16 | <b>Consumables</b>              | Ink Cartridges / Toners                                                       | 20 Sets     | -      |
| 17 |                                 | A4, A3 Papers (Matte, Glossy, Bond)                                           | 5000 Sheets | -      |
| 18 |                                 | Photo Paper Packs (Glossy, Satin)                                             | 20 Packs    | -      |
| 19 |                                 | External Storage Devices (Hard drives/USBs for project backup)                | 10          | 1:2.5  |
| 20 |                                 | Printing Chemicals (as per printer type)                                      | As needed   | -      |
| 21 | <b>Safety &amp; Maintenance</b> | Surge Protectors, Fire Extinguisher, First Aid Kit, Cleaning Kits for Devices | 5 Kits      | 1:5    |

## **MODULE III**

## DATABASE MANAGEMENT

**ISCED UNIT CODE:** 0613 554 05A

**UNIT CODE:** ICT/CU/CS/CR/05/6/MA

### Relationship to Occupational Standards

This unit addresses the unit of competency: Manage Databases

**Duration of Unit:** 210 hours

### Unit Description:

This unit covers the competencies required to demonstrate database management skills. It involves understanding database fundamentals, designing a database, using Structured Query Language, understanding design of object-oriented databases, understanding indexing and hashing and understanding database applications.

### Summary of Learning Outcomes:

By the end of the unit, the trainee should be able to:

| Learning Outcomes                      | Durations (Hours) |
|----------------------------------------|-------------------|
| 1. Database fundamentals               | 40                |
| 2. Database design                     | 30                |
| 3. Structured Query Language           | 60                |
| 4. Design of object-oriented databases | 30                |
| 5. Indexing and hashing                | 25                |
| 6. Database applications               | 25                |
| <b>TOTAL</b>                           | <b>210</b>        |

### Learning Outcomes, Content and Suggested Assessment Methods

| Learning Outcome         | Content                                                                 | Suggested Assessment Methods                                                                                   |
|--------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 1. Database fundamentals | 1.1 Definition of database<br>1.2 Database terminologies<br>1.2.1 Table | <ul style="list-style-type: none"><li>• Oral tests</li><li>• Written tests</li><li>• Practical tests</li></ul> |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                    | 1.2.2 Database engine<br>1.2.3 Records<br>1.2.4 Field<br>1.3 Reasons of using databases<br>1.4 Definition of relational model<br>1.5 Relational Modelling Concepts<br>1.5.1 Relations/tables<br>1.5.2 Attributes/Columns<br>1.5.3 Domain<br>1.5.4 Tuples/Rows<br>1.5.5 Primary Key<br>1.5.6 Foreign Key<br>1.6 Properties of a relation/table<br>1.7 Comparison of RDBMS products<br>1.7.1 Oracle<br>1.7.2 MS SQL server<br>1.7.3 My SQL<br>1.7.4 Ms Access<br>1.8 Installation of MS SQL server<br>1.9 MS SQL server interface<br>1.10 Properties of MS SQL server<br>Database<br>1.11 Prescribe RDBMS product for<br>a simulated environment<br>1.12 Database security<br>1.12.1 Definition<br>1.12.2 Access control<br>1.12.3 Authentication<br>1.12.4 Integrity control<br>1.12.5 Backup |                                                                                                                    |
| 2. Database design | 2.1 Phases of database Design<br>2.1.1 Conceptual database<br>design (ERM Modeling)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>• Oral tests</li> <li>• Written tests</li> <li>• Practical tests</li> </ul> |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                    |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                                    | <p>2.1.2 Logical database design</p> <p>2.1.3 Physical database design</p> <p>2.2 Entity modelling</p> <p>2.2.1 Components</p> <p>2.2.2 Designing Entity Model using UML (Unified Modelling Language)</p> <p>2.3 Normalisation</p> <p>2.3.1 Definition</p> <p>2.3.2 Demonstration of normalisation</p> <p>2.4 Validating model according to the requirements / specified transactions (CRUD matrix)</p>                              |                                                                                                                    |
| 3. Structured Query Language (SQL) | <p>3.1 SQL</p> <p>3.1.1 Definition</p> <p>3.1.2 Characteristics</p> <p>3.1.3 Components</p> <p>3.2 Data definition queries</p> <p>3.2.1 CREATE</p> <p>3.2.2 DROP</p> <p>3.2.3 ALTER</p> <p>3.3 Demonstration of CREATE TABLE statement</p> <p>3.4 Demonstration of CREATE TABLE constraints:</p> <p>3.4.1 PRIMARY KEY</p> <p>3.4.2 FOREIGN KEY</p> <p>3.4.3 NOT NULL</p> <p>3.4.4 CHECK</p> <p>3.4.5 UNIQUE</p> <p>3.4.6 DEFAULT</p> | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Oral tests</li> <li>• Written tests</li> </ul> |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                              |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
|                                     | <p>3.5 Editing table schema using SQL ALTER statement</p> <p>3.5.1 Adding an attribute</p> <p>3.5.2 Dropping an attribute</p> <p>3.5.3 Modifying attribute domain</p> <p>3.6 Dropping table using SQL DROP TABLE statement</p> <p>3.7 Data manipulation query statements</p> <p>3.7.1 INSERT</p> <p>3.7.2 SELECT</p> <p>3.7.3 UPDATE</p> <p>3.7.4 DELETE</p> <p>3.8 Data Manipulation Query Statements</p> <p>3.8.1 Retrieving records using SELECT statement</p> <p>3.8.2 Insertion of records using INSERT INTO statements</p> <p>3.8.3 Deleting records using DELETE statement</p> <p>3.8.4 Updating records using UPDATE. SET statement</p> <p>3.9 SQL Joins</p> <p>3.9.1 Definition of a join</p> <p>3.9.2 Types of joins</p> <p>3.10 Create and query a database from a validated ER model.</p> <p>3.11 Creating a simple join</p> |                                                                                                              |
| 4. Object oriented databases design | <p>4.1 Object oriented database</p> <p>4.1.1 Definition</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Oral</li> <li>• Written tests</li> </ul> |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                              |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
|                          | <p>4.1.2 Comparison with other types of databases</p> <p>4.2 Object oriented database concepts</p> <p>4.2.1 Classes</p> <p>4.2.2 Objects</p> <p>4.2.3 Attributes</p> <p>4.2.4 Inheritance</p> <p>4.3 Implementation of Object Oriented Database Concepts from a set of requirements</p> <p>4.4 Creation of views and triggers.</p>                                                                          |                                                                                                              |
| 5. Indexing and hashing  | <p>5.1 Indexing and hashing</p> <p>5.1.1 Definition of indexing and hashing</p> <p>5.1.2 Types of indexing</p> <p>5.1.3 Types of hashing</p> <p>5.2 Demonstration of indexing</p> <p>5.2.1 Dense index</p> <p>5.2.2 Sparse index</p> <p>5.3 Demonstration of hashing</p> <p>5.3.1 Static hashing</p> <p>5.3.2 Dynamic hashing</p> <p>5.4 Implementation of indexing and hashing in an existing database</p> | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Oral</li> <li>• Written tests</li> </ul> |
| 6. Database applications | <p>6.1 Decision support system</p> <p>6.2 Data mining</p> <p>6.3 Distributed Databases</p> <p>6.4 Features of Data warehouses</p> <p>6.5 Features of Spatial and geographical databases</p> <p>6.6 Features of Multi-media databases</p> <p>6.7 Mobility and personal databases</p>                                                                                                                         | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Oral</li> <li>• Written tests</li> </ul> |

|  |                                                  |  |
|--|--------------------------------------------------|--|
|  | 6.8 Design and implementation of data warehouses |  |
|--|--------------------------------------------------|--|

### Suggested Methods of Instruction

- Presentations and practical demonstrations by trainer;
- Guided learner activities and research to develop underpinning knowledge;
- Supervised practical database design and SQL projects
- Visiting expert from the ICT sector;
- Industrial visits

### Recommended Resources for 25 trainees

| S/No. | Category            | Item Description / Specifications                                                            | Quantity  | Recommended Ratio (Trainee: Item) |
|-------|---------------------|----------------------------------------------------------------------------------------------|-----------|-----------------------------------|
| 1     | Learning Materials  | Database Textbooks                                                                           | 25        | 1:1                               |
| 2     |                     | Printed or digital lecture notes, case studies, ERD templates, normalization guides          | 25 Sets   | 1:1                               |
| 3     |                     | SQL Cheat Sheets & Syntax Reference Guides                                                   | 25        | 1:1                               |
| 4     |                     | Access to online platforms                                                                   | 25 Logins | 1:1                               |
| 5     | Learning Facilities | Training room with 25 workstations, whiteboard, projector, and proper lighting & ventilation | 1 Room    | -                                 |



|    |                              |                                                                                 |             |       |
|----|------------------------------|---------------------------------------------------------------------------------|-------------|-------|
| 6  |                              | Smart Board or Projector with HDMI/VGA for demonstrations                       | 1           | 1:25  |
| 7  | <b>Infrastructure</b>        | High-speed Internet                                                             | 1 Setup     | -     |
| 8  |                              | Backup power (UPS and/or Generator)                                             | 1 Unit      | -     |
| 9  | <b>Tools &amp; Equipment</b> | Laptops or Desktops.                                                            | 25          | 1:1   |
| 10 |                              | RDBMS Software (MS SQL Server, MySQL, Oracle Express, PostgreSQL, MS Access)    | 25 Licenses | 1:1   |
| 11 |                              | SQL Client Tools (e.g., SQL Server Management Studio, DBeaver, MySQL Workbench) | 25          | 1:1   |
| 12 |                              | UML and ER Modeling Tools (e.g., Lucidchart, Draw.io, Visual Paradigm, StarUML) | 25          | 1:1   |
| 13 |                              | Code Editor (e.g., Visual Studio Code, Sublime Text, Notepad++)                 | 25          | 1:1   |
| 14 |                              | Virtual Machines / Docker for                                                   | 10          | 1:2.5 |

|           |                             |                                                                                                    |                  |       |
|-----------|-----------------------------|----------------------------------------------------------------------------------------------------|------------------|-------|
|           |                             | Simulated Environments (e.g., Oracle VM, VirtualBox, Docker Desktop)                               |                  |       |
| <b>15</b> | <b>Consumables</b>          | Notebooks, pens, whiteboard markers, flipcharts                                                    | 25 Sets          | 1:1   |
| <b>16</b> |                             | Printed ER diagrams, case study handouts                                                           | 25 Sets          | 1:1   |
| <b>17</b> |                             | USB drives or external hard drives                                                                 | 10               | 1:2.5 |
| <b>18</b> |                             | SQL Script Samples and Practice Datasets                                                           | Shared           | -     |
| <b>19</b> | <b>Software Licenses</b>    | Microsoft SQL Server Developer Edition / MySQL / Oracle XE (free editions or educational licenses) | 25 Installations | 1:1   |
| <b>20</b> | <b>Safety &amp; Support</b> | Anti-virus Software, Fire Extinguisher, First Aid Kit, Surge Protectors                            | 5 Kits           | 1:5   |

## WEB DESIGNING

**ISCED UNIT CODE:** 0613 554 06A

**UNIT CODE:** ICT/CU/CS/CR/10/6/MA

### Relationship to Occupational Standards

This unit addresses the unit of competency: Design Web Applications

**Duration of Unit:** 220 hours

### Unit Description:

This unit covers the competencies required to demonstrate web design skills. It involves understanding HTML basics, using HTML elements, demonstrating web page formatting, applying styles, understanding JavaScript basics, using JavaScript data types, using JavaScript functions and using JavaScript libraries.

### Summary of Learning Outcomes:

| Learning Outcomes        | Durations (Hours) |
|--------------------------|-------------------|
| 1. HTML basics           | 20                |
| 2. HTML elements         | 40                |
| 3. Web page formatting   | 25                |
| 4. Styles                | 30                |
| 5. JavaScript basics     | 25                |
| 6. JavaScript data types | 30                |
| 7. JavaScript functions  | 20                |
| 8. JavaScript libraries  | 30                |
| <b>TOTAL</b>             | <b>220</b>        |

## Learning Outcomes, Content and Suggested Assessment Methods

| Learning Outcome       | Content                                                                                                                                                                                                                                                                                                                                                              | Suggested Assessment Method                                                                                        |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1. HTML basics         | 1.1 Definition of HTML<br>1.2 HTML terminologies<br>1.2.1 Document<br>1.2.2 Stylesheet<br>1.2.3 Element<br>1.2.4 Attribute<br>1.3 Creation of HTML file<br>1.3.1 Document type declaration<br>1.3.2 Saving as .html file<br>1.4 HTML core elements<br>1.4.1 <head><br>1.4.2 <title><br>1.4.3 <body><br>1.4.4 <html><br>1.5 Integration of HTML core elements to file | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Written tests</li> <li>• Oral tests</li> </ul> |
| 2. HTML elements       | 2.1 Basic HTML elements<br>2.1.1 <p><br>2.1.2 <br><br>2.1.3 <h1><br>2.2 Addition of basic HTML elements to HTML document<br>2.3 Definition of attributes<br>2.3.1 src<br>2.3.2 alt<br>2.3.3 href<br>2.4 Addition of attributes to elements                                                                                                                           | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Written tests</li> <li>• Oral tests</li> </ul> |
| 3. Web page formatting | 3.1 Layout elements<br>3.1.1 <header><br>3.1.2 <nav>                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Written tests</li> </ul>                       |

|                          |                                                                                                                                                                                                                 |                                                                                                                    |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                          | 3.1.3 <section><br>3.1.4 <footer><br>3.2 Addition of layout elements to HTML document<br>3.3 Addition of layout element attributes to HTML document<br>3.3.1 class<br>3.3.2 id<br>3.3.3 name                    | <ul style="list-style-type: none"> <li>• Oral tests</li> </ul>                                                     |
| 4. Styles                | 4.1 Style concepts<br>4.1.1 background<br>4.1.2 padding<br>4.1.3 alignment<br>4.1.4 border<br>4.2 Application of internal styles<br>4.3 Creation of external CSS file                                           | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Written tests</li> <li>• Oral tests</li> </ul> |
| 5. JavaScript basics     | 5.1 Purpose of JavaScript<br>5.2 JavaScript syntax<br>5.3 Accessing HTML element attributes using the JavaScript Document Object Model (DOM)<br>5.4 Changing HTML element attributes using JavaScript DOM model | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Written tests</li> <li>• Oral tests</li> </ul> |
| 6. JavaScript data types | 6.1 JavaScript data types<br>6.1.1 Strings<br>6.1.2 Numbers<br>6.1.3 Booleans<br>6.2 Data type operations<br>6.2.1 Variables declarations and scope<br>6.2.2 Expressions<br>6.2.2.1 Arithmetic                  | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Written tests</li> <li>• Oral tests</li> </ul> |

|                         |                                                                                                                                                                                                                                                             |                                                                                                                    |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                         | 6.2.2.2 Boolean<br>6.2.2.3 String concatenation<br>6.3 Arrays operations<br>6.3.1 count ()<br>6.3.2 pop ()<br>6.3.3 push()                                                                                                                                  |                                                                                                                    |
| 7. JavaScript functions | 7.1 JavaScript function structure<br>7.2 Creation of JavaScript function<br>7.3 Invoking of JavaScript function<br>7.4 Returning values from functions                                                                                                      | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Written tests</li> <li>• Oral tests</li> </ul> |
| 8. JavaScript libraries | 8.1 Libraries concept<br>8.2 JQuery framework<br>8.3 Installation of JQuery<br>8.4 Referencing JQuery<br>8.5 JQuery syntax<br>8.6 JQuery events<br>8.6.1 Keyboard<br>8.6.2 Mouse<br>8.6.3 Form<br>8.6.4 Document Window<br>8.7 DOM manipulation with JQuery | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Written tests</li> <li>• Oral tests</li> </ul> |

### Suggested Methods of Instruction

- Presentations and practical demonstrations by trainer;
- Guided learner activities and research to develop underpinning knowledge;
- Supervised practical assignments and projects

### Recommended Resources for 25 trainees

| S/No. | Category            | Item Description / Specifications                                                              | Quantity    | Recommended Ratio (Trainee: Item) |
|-------|---------------------|------------------------------------------------------------------------------------------------|-------------|-----------------------------------|
| 1     | Learning Materials  | Web development textbooks covering HTML5, CSS3, JavaScript, and jQuery                         | 25          | 1:1                               |
| 2     |                     | Practical lab manuals and code snippet handouts                                                | 25 Sets     | 1:1                               |
| 3     |                     | Access to online platforms                                                                     |             | 1:1                               |
| 4     | Learning Facilities | Computer lab with 25 workstations, smart board/projector, whiteboard, ergonomic chairs & desks | 1 Room      | -                                 |
| 5     |                     | Projector / LED Screen                                                                         | 1 Unit      | 1:25                              |
| 6     | Infrastructure      | Stable internet connection                                                                     | 1 Setup     | -                                 |
| 7     |                     | Power backup system (UPS or generator)                                                         | 1 Unit      | -                                 |
| 8     | Tools & Equipment   | Laptops or Desktops                                                                            | 25          | 1:1                               |
| 9     |                     | Text editors (e.g., Visual Studio Code, Sublime Text, Notepad++)                               | 25 Installs | 1:1                               |
| 10    |                     | Web browsers                                                                                   |             | 1:1                               |
| 11    |                     | Code playgrounds (e.g., CodePen, JSFiddle, Repl.it)                                            | 25 Logins   | 1:1                               |

|    |                             |                                                                                      |             |       |
|----|-----------------------------|--------------------------------------------------------------------------------------|-------------|-------|
| 12 |                             | Git and GitHub account setup for version control and collaboration                   | 25          | 1:1   |
| 13 | <b>Software/Frameworks</b>  | Node.js & NPM (for package management and jQuery use)                                | 25 Installs | 1:1   |
| 14 |                             | jQuery library (locally or via CDN reference)                                        | Shared      | -     |
| 15 | <b>Consumable Materials</b> | Printed HTML/CSS/JavaScript cheat sheets, exercise files, notebooks, pens            | 25 Sets     | 1:1   |
| 16 |                             | Flash drives or shared cloud storage (Google Drive/OneDrive/Dropbox) for code backup | 10          | 1:2.5 |
| 17 | <b>Safety &amp; Support</b> | Surge protectors, antivirus software, fire extinguisher, cleaning supplies           | 5 Kits      | 1:5   |



## **MODULE IV**

## BASIC ELECTRONICS

**ISCED UNIT CODE:** 0613 554 07A

**UNIT CODE:** ICT/CU/CS/CC/01/6/MA

### Relationship to Occupational Standards

This unit addresses the unit of competency: Apply Basic Electronic Skills

**Duration of Unit:** 180 hours

### Unit description

This unit specifies the competencies required to apply basic electronics skills. It involves identifying electric circuits and electronic components, understanding semi-conductor theory, identifying and classifying memories, applying number systems and binary coding and identifying emerging trends in electronics.

### Summary of Learning Outcomes

| Learning Outcomes                   | Duration (hours) |
|-------------------------------------|------------------|
| 1. Electric circuits                | 20               |
| 2. Electronic components            | 25               |
| 3. Semi-conductor theory            | 50               |
| 4. Memory                           | 20               |
| 5. Number systems and binary coding | 50               |
| 6. Emerging trends in electronics   | 15               |
| <b>TOTAL</b>                        | <b>180</b>       |

## Learning Outcomes, Content and Suggested Assessment Methods

| Learning outcomes        | Content                                                                                                                                                                                                                                                                                                                  | Suggested Assessment Methods                                                                                                      |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1. Electrical circuits   | 1.1 Definition of electrical circuit.<br>1.2 Basic electrical quantities and their units<br>1.2.1 E.m.f in volts<br>1.2.2 Current in Amperes<br>1.2.3 Power in watts<br>1.2.4 Energy in joules<br>1.2.5 Resistance in ohms<br>1.3 Types of electrical circuits<br>1.3.1 Simple a.c circuits<br>1.3.2 Simple d.c circuits | <ul style="list-style-type: none"> <li>• Practical exercises</li> <li>• Written</li> <li>• Observation</li> <li>• Oral</li> </ul> |
| 2. Electronic components | 2.1 Identification of electronic components<br>2.1.1 Resistor<br>2.1.2 Capacitor<br>2.1.3 Diode<br>2.1.4 Inductor<br>2.2 Characteristic of electronic components.<br>2.3 Application of electronic components.<br>2.4 Identification of integrated circuit characteristics                                               | <ul style="list-style-type: none"> <li>• Practical exercises</li> <li>• Written</li> <li>• Observation</li> <li>• Oral</li> </ul> |
| 3. Semi-conductor theory | 3.1 Definition of semiconductor and related terms<br>3.1.1 Atom<br>3.1.2 Atomic structure<br>3.2 Description of the structure of matter                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>• Practical exercises</li> <li>• Written</li> <li>• Observation</li> <li>• Oral</li> </ul> |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                    |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
|                                     | <p>3.3 Explanation of electrons in conductors and semiconductors</p> <p>3.4 Types of semiconductors materials</p> <p>3.4.1 Silicon</p> <p>3.4.2 germanium</p> <p>3.5 Explanation of P-type and N-types materials</p> <p>3.5.1 P-type</p> <p>3.5.2 N-type</p> <p>3.6 Description of P-N junction diodes operations</p> <p>3.6.1 Forward biasing</p> <p>3.6.2 Reverse biasing</p> <p>3.7 Operations of transistors</p> <p>3.7.1 PNP type</p> <p>3.7.2 NPN type</p> |                                                                                                    |
| 4. Memory                           | <p>4.1 Definition of memory</p> <p>4.2 Classification of memories(hierarchy)</p> <p>4.2.1 RAM</p> <p>4.2.2 ROM</p> <p>4.2.3 DAM</p> <p>4.3 Types of memories</p> <p>4.3.1 Semiconductor memories</p> <p>4.3.2 Magnetic memories</p>                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Written</li> <li>• Observation</li> <li>• Oral</li> </ul> |
| 5. Number systems and binary coding | <p>5.1 Definition of number system and binary code</p> <p>5.2 Types of number systems</p> <p>5.2.1 Decimal</p> <p>5.2.2 Binary</p> <p>5.2.3 Octal</p> <p>5.2.4 Hexadecimal</p>                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>• Written</li> <li>• Observation</li> <li>• Oral</li> </ul> |

|                                   |                                                                                                                                                                                                                                                                                                                                      |                                                                                                    |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
|                                   | 5.3 Base conversion<br>5.4 Binary arithmetic<br>5.4.1 Addition<br>5.4.2 Subtraction<br>5.4.3 Multiplication<br>5.4.4 Division<br>5.5 Binary codes<br>5.5.1 8421 BCD<br>5.5.2 Excess-3<br>5.6 Represent decimal numbers in BCD<br>5.7 BCD arithmetic<br>5.7.1 Addition<br>5.7.2 Subtraction<br>5.7.3 Multiplication<br>5.7.4 Division |                                                                                                    |
| 6. Emerging trends in Electronics | 6.1 Description of emerging trends<br>6.2 Explanation of challenges of emerging trends<br>6.3 Coping with the emerging trends                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• Written</li> <li>• Observation</li> <li>• Oral</li> </ul> |

### Suggested Methods of Instruction

- Presentations and practical demonstrations by trainer;
- Guided learner activities and research to develop underpinning knowledge;
- Supervised activities and projects in a workshop;
- Visiting lecturer/trainer from the ICT sector;
- Industrial visits.

### Recommended Resources for 25 trainees

| S/No. | Category                     | Item Description / Specifications                                            | Quantity  | Trainee:Item Ratio |
|-------|------------------------------|------------------------------------------------------------------------------|-----------|--------------------|
| 1.    | <b>Learning Materials</b>    | Textbooks                                                                    | 25        | 1:1                |
| 2.    |                              | Handouts                                                                     | 25 Sets   | 1:1                |
| 3.    |                              | Access to online resources                                                   | 25 Logins | 1:1                |
| 4.    | <b>Learning Facilities</b>   | Electronics lab/classroom with benches, storage, ventilation                 | 1 Room    | Shared             |
| 5.    |                              | Whiteboard, projector/smart board                                            | 1         | 1:25               |
| 6.    | <b>Infrastructure</b>        | Reliable power supply and backup (UPS/generator)                             | 1 Setup   | Shared             |
| 7.    | <b>Consumable Materials</b>  | Resistors, capacitors, diodes, ICs, wires, breadboards, soldering wire       | Assorted  | Shared             |
| 8.    |                              | Notebooks, pens, datasheets, handouts                                        | 25 Sets   | 1:1                |
| 9.    | <b>Tools &amp; Equipment</b> | Multimeters (digital)                                                        | 10        | 1:2.5              |
| 10.   |                              | Breadboards                                                                  | 25        | 1:1                |
| 11.   |                              | Soldering stations with accessories                                          | 10        | 1:2.5              |
| 12.   |                              | Power supply units (0–30V adjustable)                                        | 10        | 1:2.5              |
| 13.   |                              | Oscilloscopes (for circuit testing and signal analysis)                      | 5         | 1:5                |
| 14.   |                              | Component kits (resistors, capacitors, diodes, transistors, ICs)             | 25 Kits   | 1:1                |
| 15.   |                              | Desktop/laptop with simulation software (e.g., Multisim, Proteus, Tinkercad) | 25        | 1:1                |

## FUNDAMENTALS OF PROGRAMMING

**ISCED UNIT CODE:** 0613 554 08A

**UNIT CODE:** ICT/CU/CS/CR/04/6/MA

### Relationship to Occupational Standards

This unit addresses the unit of competency: Apply Fundamentals of Programming

**Duration of Unit:** 200 hours

### Unit Description

This unit covers the competencies required to understand fundamentals of programming. It involves understanding programming concepts, understanding the Java environment, performing data operations, using control structures, using methods and understanding Object Oriented programming.

### Summary of Learning Outcomes:

| Learning Outcomes              | Duration (hours) |
|--------------------------------|------------------|
| 1. Programming Concepts        | 20               |
| 2. Java environment            | 30               |
| 3. Data operations             | 35               |
| 4. Control Structures          | 35               |
| 5. Use methods                 | 40               |
| 6. Object Oriented Programming | 40               |
| <b>TOTAL</b>                   | <b>200</b>       |

### Learning Outcomes, Content and Suggested Assessment Methods

| Learning Outcome        | Content                                                          | Suggested Assessment Methods                                                           |
|-------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 1. Programming Concepts | 1.1 Concepts of programming<br>1.2 Phases of program development | <ul style="list-style-type: none"><li>• Practical tests</li><li>• Oral tests</li></ul> |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                    |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                            | 1.2.1 Establish program requirements<br>1.2.2 Design a program<br>1.2.3 Coding<br>1.2.4 Code test and debug<br>1.2.5 Document<br>1.2.6 Maintain<br>1.3 Key terms used in programming<br>1.3.1 Algorithm<br>1.3.2 Source code<br>1.3.3 Executable<br>1.3.4 Compiling<br>1.3.5 Debugging<br>1.4 Types of code<br>1.4.1 Source code<br>1.4.2 Object code<br>1.4.3 Machine code<br>1.5 Translators used in programming<br>1.5.1 Compiler<br>1.5.2 Interpreter<br>1.5.3 Assembler<br>1.6 OOP fundamental concepts | <ul style="list-style-type: none"> <li>• Written tests</li> </ul>                                                  |
| <b>2. Java Environment</b> | 2.1 Installation of Java<br>2.1.1 Download Java for Windows<br>2.1.2 Install JDK<br>2.1.3 Set the Environment variables<br>2.2 Java Programming environment<br>2.2.1 Downloading Eclipse IDE<br>2.2.2 Setting up Eclipse IDE<br>2.2.3 Launching Eclipse IDE<br>2.3 Features of Java<br>2.4 Java syntax<br>2.4.1 Case Sensitivity                                                                                                                                                                             | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Oral tests</li> <li>• Written tests</li> </ul> |



|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                    |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                    | 2.4.2 Class names<br>2.4.3 Method names<br>2.4.4 Program file name<br>2.4.5 Public static void main<br>2.4.6 Identifiers<br>2.4.7 Modifiers<br>2.4.8 Variables<br>2.4.9 Java Arrays<br>2.4.10 Java Enums<br>2.4.11 Java Keywords                                                                                                                                                                                                                                                                                                                                             |                                                                                                                    |
| 3. Data Operations | 3.1 Java Data Types 3.1.1 Integer<br>3.1.2 Float<br>3.1.3 Strings<br>3.1.4 Boolean<br>3.2 Java statements 3.2.1 Expression Statements<br>3.2.2 Declaration Statements<br>3.2.3 Control-flow statements<br>3.3 Variables and Constants 3.3.1 Local Variables<br>3.3.2 Class Variables<br>3.3.3 Instance Variables<br>3.3.4 Integer constants<br>3.3.5 Real Constants<br>3.3.6 Single character constants<br>3.3.7 String constants<br>3.4 Java Data operations 3.4.1 Variable assignment<br>3.4.2 Variable reading<br>3.4.3 Variable arithmetic<br>3.4.4 Object Instantiation | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Oral tests</li> <li>• Written tests</li> </ul> |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                    |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                          | 3.5 Java Program to perform an operation<br>3.5.1 Area of a circle<br>3.5.2 Solve Quadratic equations<br>3.5.3 Calculate compound interest                                                                                                                                                                                                                                                                                                                                             |                                                                                                                    |
| 4. Use Control structure | 4.1 Java Control Statements<br>4.1.1 Decision making statements<br>4.1.2 Looping statements<br>4.1.3 Branching statements<br>4.2 Uses of different control statements in Java<br>4.2.1 Decision making statements<br>4.2.1.1 If then<br>4.2.1.2 If then else<br>4.2.1.3 Switch<br>4.2.2 Looping statements<br>4.2.2.1 for<br>4.2.2.2 while<br>4.2.2.3 do while<br>4.2.3 Branching statements<br>4.2.3.1 break<br>4.2.3.2 Continue<br>4.3 Creation of programs using control statements | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Oral tests</li> <li>• Written tests</li> </ul> |
| 5. Use Methods           | 5.1 Java Methods<br>5.1.1 Definition<br>5.1.2 Structure<br>5.2 Demonstration of methods<br>5.2.1 Creating Methods<br>5.2.2 Method calling<br>5.2.3 Void keyword<br>5.2.4 Passing parameters by value                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Oral tests</li> <li>• Written tests</li> </ul> |

|                                |                                                                                                                                                                                                                                                                                                                               |                                                                                                                    |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                                | 5.2.5 Method overloading<br>5.2.6 Using command line arguments<br>5.2.7 This keyword<br>5.2.8 Variable arguments<br>5.2.9 The finalize () method<br>5.3 Creation programs to implement methods                                                                                                                                |                                                                                                                    |
| 6. Object Oriented Programming | 6.1 Object oriented programming concepts<br>6.1.1 Inheritance<br>6.1.2 Encapsulation<br>6.1.3 Abstraction<br>6.1.4 Polymorphism<br>6.2 Classes<br>6.2.1 Declaring attributes<br>6.2.2 Creating Methods<br>6.3 Objects<br>6.3.1 Creating objects<br>6.3.2 Calling methods<br>6.4 Creation of programs to implement inheritance | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Oral tests</li> <li>• Written tests</li> </ul> |

### Suggested Methods of Instruction

- Presentations and practical demonstrations by trainer;
- Guided learner activities and research to develop underpinning knowledge;
- Supervised practical assignments and projects;
- Visiting lecturer/expert from the ICT sector;
- Industrial visits.

### Recommended Resources for 25 trainees

| S/No. | Category                     | Item Description / Specifications                                             | Quantity    | Trainee:Item Ratio |
|-------|------------------------------|-------------------------------------------------------------------------------|-------------|--------------------|
| 1.    | <b>Learning Materials</b>    | Java Programming textbooks                                                    | 25          | 1:1                |
| 2.    |                              | Java handouts, syntax reference sheets, sample programs                       | 25 Sets     | 1:1                |
| 3.    |                              | Access to online platforms                                                    | 25 Logins   | 1:1                |
| 4.    | <b>Learning Facilities</b>   | Computer lab/classroom with 25 workstations, whiteboard, projector            | 1 Room      | Shared             |
| 5.    | <b>Infrastructure</b>        | Stable internet connection                                                    | 1 Setup     | Shared             |
| 6.    | <b>Tools &amp; Equipment</b> | Laptops/Desktops                                                              | 25          | 1:1                |
| 7.    |                              | Java Development Kit (JDK) installed                                          | 25 Installs | 1:1                |
| 8.    |                              | Eclipse IDE or IntelliJ IDEA Community Edition installed                      | 25 Installs | 1:1                |
| 9.    |                              | Code playgrounds (e.g., Repl.it, JDoodle, OnlineGDB for Java)                 | 25 Logins   | 1:1                |
| 10.   |                              | Git & GitHub accounts for version control and code collaboration              | 25 Accounts | 1:1                |
| 11.   | <b>Consumable Materials</b>  | Notebooks, pens, printed exercises, test papers                               | 25 Sets     | 1:1                |
| 12.   |                              | USB drives or cloud storage access (Google Drive, OneDrive, etc.)             | 10          | 1:2.5              |
| 13.   | <b>Safety &amp; Support</b>  | Antivirus software, surge protectors, fire extinguishers, basic first aid kit | 5 Kits      | 1:5                |

## WORK ETHICS AND PRACTICES

**ISCED UNIT CODE:** 0417 441 02A

**TVET CDACC UNIT CODE:** IT/CU/ICTA/BC/01/5/MA

**Duration of Unit:** 40 hours

### Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply work ethics and practices.

### Unit Description

This unit covers competencies required to effectively apply work ethics and practices. It involves the ability to: apply self-management skills, promote ethical work practices and values, promote teamwork, maintain professional and personal development, apply problem solving and promote customer care.

### Summary of Learning Outcomes

| LEARNING OUTCOMES                                 | DURATION (HOURS) |
|---------------------------------------------------|------------------|
| 1. Apply self-management skills                   | 10               |
| 2. Promote ethical practices and values           | 10               |
| 3. Promote teamwork                               | 5                |
| 4. Maintain professional and personal development | 5                |
| 5. Apply problem-solving skills                   | 5                |
| 6. Promote customer care.                         | 5                |
| <b>TOTAL:</b>                                     | <b>40</b>        |

## Learning Outcomes, Content, and Suggested Assessment Methods

| Learning Outcome                             | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Suggested Assessment Methods                                                                                                                                                                                                     |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Apply self-management skills              | 1.1. Self-awareness<br>1.2. Formulating personal vision, mission, and goals<br>1.3. Healthy lifestyle practices<br>1.4. Strategies for overcoming work challenge<br>1.5. Emotional intelligence<br>1.6. Coping with Work Stress.<br>1.7. Assertiveness versus aggressiveness and passiveness<br>1.8. Developing and maintaining high self-esteem<br>1.9. Developing and maintaining positive self-image<br>1.10. Time management<br>1.11. Setting performance targets<br>1.12. Monitoring and evaluating performance targets | <ul style="list-style-type: none"> <li>• Observation</li> <li>• Written assessment</li> <li>• Oral assessment</li> <li>• Third party reports</li> <li>• Portfolio of evidence</li> <li>• Project</li> <li>• Practical</li> </ul> |
| 2. Promote ethical work practices and values | 2.1 Integrity<br>2.2 Core Values, ethics and beliefs<br>2.3 Patriotism<br>2.4 Professionalism<br>2.5 Organizational codes of conduct<br>2.6 Industry policies and procedures                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Observation</li> <li>• Written assessment</li> <li>• Oral assessment</li> <li>• Third party reports</li> <li>• Portfolio of evidence</li> <li>• Project</li> <li>• Practical</li> </ul> |

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                  |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.Promote Teamwork                                | 3.1 Types of teams<br>3.2 Team building<br>3.3 Individual responsibilities in a team<br>3.4 Determination of team roles and objectives<br>3.5 Team parameters and relationships<br>3.6 Benefits of teamwork<br>3.7 Qualities of a team player<br>3.8 Leading a team<br>3.9 Team performance and evaluation<br>3.10 Conflicts and conflict resolution<br>3.11 Gender and diversity mainstreaming<br>3.12 Developing Healthy workplace relationships<br>3.13 Adaptability and flexibility<br>3.14 Coaching and mentoring skills | <ul style="list-style-type: none"> <li>• Observation</li> <li>• Written assessment</li> <li>• Oral assessment</li> <li>• Third party reports</li> <li>• Portfolio of evidence</li> <li>• Project</li> <li>• Practical</li> </ul> |
| 4. Maintain professional and personal development | 4.1 Personal vs professional development and growth<br>4.2 Avenues for professional growth<br>4.3 Recognizing career advancement<br>4.4 Training and career opportunities<br>4.5 Assessing training needs                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>• Observation</li> <li>• Written assessment</li> <li>• Oral assessment</li> <li>• Third party reports</li> <li>• Portfolio of evidence</li> <li>• Project</li> <li>• Practical</li> </ul> |

|                                 |                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | 4.6 Mobilizing training resources<br>4.7 Licenses and certifications for professional growth and development<br>4.8 Pursuing personal and organizational goals<br>4.9 Managing work priorities and commitments<br>4.10 Dynamism and on-the-job learning |                                                                                                                                                                                                                                  |
| 5. Apply Problem-solving skills | 5.1 Causes of problems<br>5.2 Methods of solving problems<br>5.3 Problem-solving process<br>5.4 Decision making<br>5.5 Creative thinking and critical thinking process in development of innovative and practical solutions                             | <ul style="list-style-type: none"> <li>• Observation</li> <li>• Written assessment</li> <li>• Oral assessment</li> <li>• Third party reports</li> <li>• Portfolio of evidence</li> <li>• Project</li> <li>• Practical</li> </ul> |
| 6.Promote Customer Care         | 6.1 Identifying customer needs<br>6.2 Qualities of good customer service<br>6.3 Customer feedback methods<br>6.4 Resolving customer concerns<br>6.5 Customer outreach programs<br>6.6 Customer retention                                                | <ul style="list-style-type: none"> <li>• Observation</li> <li>• Written assessment</li> <li>• Oral assessment</li> <li>• Third party reports</li> <li>• Portfolio of evidence</li> <li>• Project</li> <li>• Practical</li> </ul> |

### **Suggested Methods of Instruction**

- Instructor lead facilitation of theory using active learning strategies.
- Demonstrations
- Simulation/Role play
- Group Discussion



- Presentations
- Projects
- Case studies
- Assignments

#### Recommended Resources for 25 Trainees

| S/No.    | Category/Item                                   | Description / Specifications                              | Quantity    | Recommended Ratio (Trainee:Item) |
|----------|-------------------------------------------------|-----------------------------------------------------------|-------------|----------------------------------|
| <b>A</b> | <b>Learning Materials</b>                       |                                                           |             |                                  |
| 1        | Textbooks                                       | AI, Python programming, Machine Learning guides           | 5 pcs       | 5:1                              |
| 2        | PowerPoint Presentations                        | AI concepts, Python syntax, ML algorithms (trainer's use) | For trainer | N/A                              |
| 3        | Assorted Whiteboard Markers                     | Multi-color, dry erase markers                            | 2 packets   | For trainer                      |
| 4        | e-Didactics                                     | Digital content and online resources                      | For trainer | N/A                              |
| 5        | Flashcards                                      | AI/ML/Python key term flashcards                          | 25 sets     | 1:1                              |
| 6        | Flip Charts                                     | For brainstorming and instruction display                 | 2 pads      | 1:13                             |
| 7        | Whiteboard                                      | Magnetic or non-magnetic instructional board              | 1 board     | Shared                           |
| <b>B</b> | <b>Learning Facilities &amp; Infrastructure</b> |                                                           |             |                                  |
| 8        | Lecture / Theory Room                           | Equipped with chairs, desks, ventilation                  | 1 room      | 25:1                             |

|          |                             |                                                     |              |        |
|----------|-----------------------------|-----------------------------------------------------|--------------|--------|
| <b>C</b> | <b>Consumable Materials</b> |                                                     |              |        |
| 9        | Printing Papers             | A4 ream, for notes, assignments                     | 1 ream       | 1:20   |
| 10       | Toners                      | For laserjet printers                               | 2 pcs        | 13:1   |
| 11       | Internet Connection         | Stable broadband/Wi-Fi for installations & research | 1 connection | Shared |
| <b>D</b> | <b>Tools and Equipment</b>  |                                                     |              |        |
| 12       | Projector                   | Digital projector for demonstrations                | 1 unit       | 25:1   |
| 13       | Printers                    | Laserjet or inkjet, network enabled                 | 4 units      | 6:1    |
| 14       | Computers / Mobile Phones   | With Python installed (desktop/laptop preferred)    | 25 pcs       | 1:1    |

## **MODULE V**

## ENTREPRENEURIAL SKILLS

**ISCED UNIT CODE:** 0413 441 03A

**TVET CDACC UNIT CODE:** ICT/CU/ICTA/BC/02/5/MA

**Duration of unit:** 40 hours

### Relationship to occupational standards

This unit addresses the unit of competency: Apply Entrepreneurial skills.

### Unit Description:

This unit covers the competencies required to demonstrate an understanding of entrepreneurship. It involves the ability to: apply financial literacy, apply entrepreneurial concepts, identify entrepreneurship opportunities, apply business legal aspects, innovate business strategies, and develop business plans.

### Summary of Learning Outcomes

| Learning Outcomes                          | Duration (Hours) |
|--------------------------------------------|------------------|
| 1. Apply financial literacy                | 5                |
| 2. Apply the entrepreneurial concept       | 5                |
| 3. Identify entrepreneurship opportunities | 5                |
| 4. Apply business legal aspects            | 10               |
| 5. Innovate Business Strategies            | 5                |
| 6. Develop business plan                   | 10               |
| <b>TOTAL</b>                               | <b>40</b>        |

## Learning Outcomes, Content and Suggested Assessment Methods

| Learning Outcome                 | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Suggested Assessment Methods                                                                                                                                                                    |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Apply financial literacy      | 1.1 Personal finance management<br>1.2 Balancing between needs and wants<br>1.3 Budget Preparation<br>1.4 Savings management<br>1.5 Factors to consider when deciding where to save<br>1.6 Debt management<br>1.7 Factors to consider before taking a loan<br>1.8 Investment decisions<br>1.9 Types of investments<br>1.10 Factors to consider when investing money<br>1.11 Insurance services<br>1.11.1 Insurance products available in the market<br>1.11.2 Insurable risks | <ul style="list-style-type: none"> <li>• Observation</li> <li>• Project</li> <li>• Written assessment</li> <li>• Oral assessment</li> <li>• Third party report</li> <li>• Interviews</li> </ul> |
| 2. Apply entrepreneurial concept | 2.1 Difference between Entrepreneurs and Business persons<br>2.2 Types of entrepreneurs<br>2.3 Ways of becoming an entrepreneur<br>2.4 Characteristics of Entrepreneurs<br>2.5 Salaried employment and self-employment<br>2.6 Requirements for entry into self-employment<br>2.7 Roles of an Entrepreneur in an enterprise                                                                                                                                                    | <ul style="list-style-type: none"> <li>• Observation</li> <li>• Project</li> <li>• Written assessment</li> <li>• Oral assessment</li> <li>• Third party report</li> <li>• </li> </ul>           |

|                                            |                                                                                                                                                                                                                                        |                                                                                                                                                                           |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | 2.8 Contributions of Entrepreneurship                                                                                                                                                                                                  |                                                                                                                                                                           |
| 3. Identify entrepreneurship opportunities | 3.1 Sources of business ideas<br>3.2 Factors to consider when evaluating business opportunity<br>3.3 Business life cycle                                                                                                               | <ul style="list-style-type: none"> <li>• Observation</li> <li>• Project</li> <li>• Written assessment</li> <li>• Oral assessment</li> <li>• Third party report</li> </ul> |
| 4. Apply business legal aspects            | 4.1 Forms of business ownership<br>4.2 Business registration and licensing processing<br>4.3 Types of contracts and agreements<br>4.4 Employment laws<br>4.5 Taxation laws                                                             | <ul style="list-style-type: none"> <li>• Observation</li> <li>• Project</li> <li>• Written assessment</li> <li>• Oral assessment</li> <li>• Third party report</li> </ul> |
| 5. Innovate business Strategies            | 5.1 Creativity in business<br>5.2 Innovative business strategies<br>5.3 Entrepreneurial Linkages<br>5.4 ICT in business growth and development                                                                                         | <ul style="list-style-type: none"> <li>• Observation</li> <li>• Project</li> <li>• Written assessment</li> <li>• Oral assessment</li> <li>• Third party report</li> </ul> |
| 6. Develop Business Plan                   | 6.1 Business description<br>6.2 Marketing plan<br>6.3 Organizational/Management plan<br>6.4 Production/operation plan<br>6.5 Financial plan<br>6.6 Executive summary<br>6.7 Business plan presentation<br>6.8 Business idea incubation | <ul style="list-style-type: none"> <li>• Observation</li> <li>• Written assessment</li> <li>• Project</li> <li>• Oral assessment</li> <li>• Third party report</li> </ul> |

### Suggested Methods of Instruction

- Direct instruction with active learning strategies

- Project (Business plan)
- Case studies
- Field trips
- Group Discussions
- Demonstration
- Question and answer
- Problem solving
- Experiential
- Team training
- Guest speakers

#### **Recommended Resources for 25 Trainees**

| <b>S/No.</b> | <b>Category/Item</b>      | <b>Description / Specifications</b>                     | <b>Quantity</b> | <b>Recommended Ratio (Trainee: Item)</b> |
|--------------|---------------------------|---------------------------------------------------------|-----------------|------------------------------------------|
| <b>A</b>     | <b>Learning Materials</b> |                                                         |                 |                                          |
| 1            | Textbooks                 | Entrepreneurship/Business Studies books                 | 5 pcs           | 5:1                                      |
| 2            | Business Plan Templates   | Printed/soft templates for practice and assessment      | 5 pcs           | 5:1                                      |
| 3            | Business Journals         | Latest editions, case studies and trends                | 5 pcs           | 5:1                                      |
| 4            | Newspapers and Handouts   | Business sections, financial reports, industry trends   | For trainer     | Shared                                   |
| 5            | PowerPoint Presentations  | Business plan, marketing, finance, etc. (trainer's use) | For trainer     | N/A                                      |

|          |                                                 |                                               |              |             |
|----------|-------------------------------------------------|-----------------------------------------------|--------------|-------------|
| 6        | Assorted Whiteboard Markers                     | Multi-colour, dry erase markers               | 2 packets    | For trainer |
| 7        | e-Didactics                                     | Digital lessons, videos, online case studies  | For trainer  | N/A         |
| 8        | Flashcards                                      | Key business terms and definitions            | 25 sets      | 1:1         |
| 9        | Flip Charts                                     | Brainstorming, strategy development           | 2 pads       | 1:13        |
| 10       | Whiteboard                                      | Instructional board                           | 1 board      | Shared      |
| <b>B</b> | <b>Learning Facilities &amp; Infrastructure</b> |                                               |              |             |
| 11       | Lecture / Theory Room                           | With seating, ventilation, and basic AV setup | 1 room       | 25:1        |
| <b>C</b> | <b>Consumable Materials</b>                     |                                               |              |             |
| 12       | Printing Papers                                 | A4 papers for plans, worksheets, assessments  | 1 ream       | 1:20        |
| 13       | Toners                                          | For printers used by trainees                 | 2 pcs        | 13:1        |
| 14       | Internet Connection                             | Stable connection for research, e-learning    | 1 connection | Shared      |
| <b>D</b> | <b>Tools and Equipment</b>                      |                                               |              |             |



|    |                         |                                                       |          |      |
|----|-------------------------|-------------------------------------------------------|----------|------|
| 15 | Projectors              | Digital projection for lectures and presentations     | 1 unit   | 25:1 |
| 16 | Printers                | Shared for document output and plan printing          | 4 units  | 6:1  |
| 17 | Computers / Smartphones | For business simulations, research, and presentations | 25 units | 1:1  |

## ALGORITHMS AND DATA STRUCTURES

**ISCED UNIT CODE:** 0613 554 09A

**UNIT CODE:** ICT/CU/CS/CR/09/6/MA

### Relationship to Occupational Standards

This unit addresses the unit of competency: Create Algorithms and Data Structures

**Duration of Unit:** 190 hours

### Unit Description

This unit covers the competencies required to understand algorithms and data structure. It involves Understand fundamental principles of algorithms understanding fundamental concepts of data structures, linked lists, stacks and queues, search techniques and sorting techniques

### Summary of Learning Outcomes

| Learning Outcomes                          | Duration (Hours) |
|--------------------------------------------|------------------|
| 1. Fundamental principles of algorithms    | 30               |
| 2. Fundamental concepts of data structures | 30               |
| 3. Linked lists                            | 40               |
| 4. Stacks and Queues                       | 30               |
| 5. Search Techniques                       | 25               |
| 6. Sorting Techniques                      | 35               |
| <b>TOTAL</b>                               | <b>190</b>       |

## Learning Outcomes, Content and Suggested Assessment Methods

| Learning Outcome                           | Content                                                                                                                                                                                                                                                                                                                                                                 | Suggested Assessment Method                                                                                        |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1. Fundamental principles of algorithms    | 1.1 Algorithm principles<br>1.2 Characteristics of an Algorithm<br>1.3 Principles of algorithm writing<br>1.4 Algorithm Analysis<br>1.5 Complexities of algorithms<br>1.5.1 Space<br>1.5.2 Time<br>1.6 Greedy algorithms are outlined<br>1.6.1 Counting coins<br>1.7 Divide and conquer algorithms<br>1.7.1 Divide /break<br>1.7.2 Conquer/solve<br>1.7.3 Merge/combine | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |
| 2. Fundamental concepts of data structures | 2.1 Key concepts in data structures<br>2.1.1 Data<br>2.1.2 Object<br>2.1.3 Data type<br>2.2 Explanation of Arrays<br>2.3 Array insertion operations<br>2.3.1 At the beginning<br>2.3.2 At the given index<br>2.3.3 After the given index<br>2.3.4 Before the given index<br>2.4 Array delete, search and update<br>2.5 Demonstration of array operations                | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |
| 3. Linked lists                            | 3.1 Linked lists<br>3.1.1 Linked lists representation<br>3.1.2 Types of linked lists<br>3.2 Doubly linked lists                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |

|                       |                                                                                                                                                                                                                                                                                      |                                                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                       | 3.2.1 Representation<br>3.2.2 Basic operations<br>3.3 Circular linked lists<br>3.3.1 Representation<br>3.3.2 Basic operations<br>3.4 Demonstration of basic operations for the various linked lists using C++<br>3.4.1 Insertion<br>3.4.2 Deletion<br>3.4.3 Reverse<br>3.4.4 Display |                                                                                                                    |
| 4. Stacks and Queues  | 4.1 Definition of Stacks<br>4.2 Representation of stacks<br>4.3 Basic operations<br>4.3.1 Pop<br>4.3.2 Push<br>4.4 Definition of queues<br>4.5 Representation of queues<br>4.6 Basic operations<br>4.6.1 Enqueue<br>4.6.2 Dequeue<br>4.7 Demonstration of stack and queues using C++ | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |
| 5. Search Techniques  | 5.1 Definition of search<br>5.2 Explanation of Linear Search<br>5.3 Explanation of Binary Search<br>5.4 Demonstration of linear search and binary search using C++                                                                                                                   | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |
| 6. Sorting Techniques | 6.1 Definition of Sorting<br>6.2 Categories of sorting<br>6.2.1 Stable and not stable sorting                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Oral tests</li> <li>• Practical tests</li> </ul> |

|  |                                                                                                                                                                                                                                                   |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | 6.2.2 Adaptive and Non-Adaptive Sorting Algorithm<br>6.2.3 In place and not in place<br>6.3 Types of Sorting algorithms<br>6.3.1 Bubble sort<br>6.3.2 Insertion sort<br>6.3.3 Selection sort<br>6.4 Demonstration of sorting algorithms using C++ |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

### Suggested Methods of Instruction

- Presentations by trainer;
- Guided learner activities and research to develop underpinning knowledge;
- Supervised practical assignments
- Visiting expert from the ICT sector;
- Industrial visits

### Recommended Resources for 25 Trainees

| S/No. | Category            | Item Description / Specifications                                     | Quantity  | Trainee:Item Ratio |
|-------|---------------------|-----------------------------------------------------------------------|-----------|--------------------|
| 1     | Learning Materials  | Textbooks and handouts on algorithms, data structures, and C++ syntax | 25        | 1:1                |
|       |                     | Algorithm analysis cheat sheets and complexity tables                 | 25 Sets   | 1:1                |
|       |                     | Access to online platforms                                            | 25 Logins | 1:1                |
| 2     | Learning Facilities | Computer lab/classroom with 25 workstations, whiteboard, projector    | 1 Room    | Shared             |
| 3     | Infrastructure      | Stable internet connection                                            | 1 Setup   | Shared             |

|          |                              |                                                                         |             |       |
|----------|------------------------------|-------------------------------------------------------------------------|-------------|-------|
| <b>4</b> | <b>Tools &amp; Equipment</b> | Laptops/desktops                                                        | 25          | 1:1   |
|          |                              | Code editors (Code::Blocks, Visual Studio Code with C++ extension)      | 25 Installs | 1:1   |
|          |                              | Online compilers                                                        | 25 Logins   | 1:1   |
|          |                              | Git & GitHub for version control and practice                           | 25 Accounts | 1:1   |
| <b>5</b> | <b>Consumable Materials</b>  | Printed exercises, pens, notebooks                                      | 25 Sets     | 1:1   |
|          |                              | Flash drives or shared cloud storage (Google Drive, OneDrive, etc.)     | 10          | 1:2.5 |
| <b>6</b> | <b>Support &amp; Safety</b>  | Antivirus software, surge protectors, basic electrical safety equipment | 5 Kits      | 1:5   |

## MATHEMATICS FOR COMPUTER SCIENCE

**ISCED UNIT CODE:** 0613 554 10A

**UNIT CODE:** ICT/CU/CS/CR/03/6/MA

### Relationship to Occupational Standards

This unit addresses the unit of competency: Apply Mathematics for Computer Science

**Duration of Unit:** 170 hours

### Unit description

This unit specifies the competencies required to understanding linear algebra, understanding Boolean algebra, understanding set theory, understanding calculus and understanding probability and statistics.

### Summary of Learning Outcomes

| Learning Outcomes             | Duration (Hours) |
|-------------------------------|------------------|
| 1. Linear Algebra             | 30               |
| 2. Boolean Algebra            | 40               |
| 3. Set Theory                 | 30               |
| 4. Calculus                   | 30               |
| 5. Probability and Statistics | 40               |
| <b>TOTAL</b>                  | <b>170</b>       |

## Learning Outcomes, Content and Suggested Assessment Methods

| Learning Outcome   | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Suggested Assessment Methods                                                                                       |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1. Linear Algebra  | 1.1 Linear Equations<br>1.2 Solving linear equations<br>1.2.1 Methods of solving<br>1.2.2 Formation<br>1.3 Vectors<br>1.3.1 Definition<br>1.3.2 Types<br>1.4 Vector operations<br>1.4.1 Addition<br>1.4.2 Subtraction<br>1.4.3 Multiplication<br>1.4.4 Scalar<br>1.4.5 Dot product<br>1.5 Matrices<br>1.5.1 Definition<br>1.5.2 Types<br>1.5.3 Determinant<br>1.5.4 Application<br>1.6 Matrix operations<br>1.6.1 Addition<br>1.6.2 Scalar multiplication<br>1.6.3 Transposition<br>1.7 Inverse of square matrix | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Oral tests</li> <li>• Written tests</li> </ul> |
| 2. Boolean Algebra | 2.1 Boolean algebra<br>2.1.1 Definition of Boolean algebra<br>2.1.2 Uses of Boolean algebra<br>2.2 Boolean operations<br>2.2.1 Boolean value                                                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Oral tests</li> <li>• Written tests</li> </ul> |



|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
|               | 2.2.2 Boolean function<br>2.2.3 Digital logic<br>2.3 Basic Boolean operations<br>2.3.1 AND<br>2.3.2 OR<br>2.3.3 NOT<br>2.4 Secondary operations<br>2.4.1 NAND<br>2.4.2 NOR<br>2.4.3 EX-OR<br>2.4.4 EX-NOR<br>2.5 Writing Boolean Expressions<br>2.5.1 Order of basic operations<br>2.5.2 Symbols<br>2.6 Simplification of Boolean expressions<br>2.6.1 Using algebraic functions<br>2.6.2 Using Truth tables<br>2.6.3 Using Karnaugh Maps<br>2.7 Boolean Laws and Theorems<br>2.7.1 AND law<br>2.7.2 OR law<br>2.7.3 Inversion law<br>2.7.4 Commutative<br>2.7.5 Associative<br>2.7.6 Distributive<br>2.7.7 De-Morgan's Theorems<br>2.8 Simplification (Reduction)<br>Rules for Boolean expressions |                                                                     |
| 3. Set Theory | 3.1 Sets Theory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>• Practical tests</li> </ul> |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
|             | 3.1.1 Definition of a Set<br>3.1.2 Characteristics of sets<br>3.2 Methods of Set representation<br>3.2.1 Statement form<br>3.2.2 Tabular form<br>3.2.3 Set builder notation<br>3.3 Cardinality of a set<br>3.4 Types of sets<br>3.4.1 Finite<br>3.4.2 Infinite<br>3.4.3 Subset<br>3.4.4 Universal<br>3.4.5 Proper<br>3.4.6 Singleton set<br>3.5 Venn Diagrams<br>3.6 Set Operations<br>3.6.1 Set Union<br>3.6.2 Set Intersection<br>3.6.3 Set Difference<br>3.6.4 Complement of Set<br>3.6.5 Cartesian Product | <ul style="list-style-type: none"> <li>• Oral tests</li> <li>• Written tests</li> </ul>            |
| 4. Calculus | 4.1 Functions<br>4.1.1 Definition of function<br>4.1.2 Domain<br>4.1.3 Range<br>4.1.4 Linear functions<br>4.1.5 Power functions<br>4.1.6 Evaluation<br>4.2 Graphing of functions<br>4.2.1 Intercepts<br>4.2.2 Limits<br>4.3 Differential calculus                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Oral</li> <li>• Observation</li> <li>• Written</li> </ul> |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                    |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                               | 4.3.1 Rate of change<br>4.3.2 Rules of derivatives<br>4.3.3 Optimization<br>4.4 First and second order differential equations<br>4.5 Integral calculus<br>4.5.1 Definite<br>4.5.2 Indefinite<br>4.6 Techniques of integration<br>4.6.1 By parts<br>4.6.2 Reserve chain rule<br>4.6.3 u-substitution                                                                                                                              |                                                                                                                    |
| 5. Probability and Statistics | 5.1 Key terminologies in probability<br>5.1.1 Samples spaces<br>5.1.2 events<br>5.1.3 sets<br>5.1.4 outcomes<br>5.2 Probability axioms and counting problems<br>5.3 Permutations and combinations<br>5.4 Conditional probability and multiplication rule<br>5.5 Data representation techniques<br>5.5.1 Histogram<br>5.5.2 Pie charts<br>5.5.3 Scatter plot<br>5.5.4 Bar graph<br>5.6 Measures of central tendency<br>5.6.1 Mean | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Oral tests</li> <li>• Written tests</li> </ul> |

|  |                                                                                                                                                                            |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | 5.6.2 Mode<br>5.6.3 Median<br>5.7 Measures of spread<br>5.7.1 Variance<br>5.7.2 Standard deviation<br>5.8 Measure of Location<br>5.8.1 Quartile<br><b>5.8.2 Percentile</b> |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

### Suggested Methods of Instruction

- Presentations by trainer;
- Guided learner activities and research to develop underpinning knowledge;
- Visiting lecturer/trainer from the Mathematics field.
- Industrial visits

### Recommended Resources for 25 Trainees

| S/No. | Category                     | Item Description / Specifications                                             | Quantity  | Trainee:Item Ratio |
|-------|------------------------------|-------------------------------------------------------------------------------|-----------|--------------------|
| 1.    | <b>Learning Materials</b>    | Mathematics textbooks and handouts                                            | 25        | 1:1                |
| 2.    |                              | Formula sheets, solved examples, practice question booklets                   | 25 Sets   | 1:1                |
| 3.    |                              | Access to math learning platforms                                             | 25 Logins | 1:1                |
| 4.    |                              | Graphing calculator reference guides                                          | 25        | 1:1                |
| 5.    | <b>Learning Facilities</b>   | Well-lit classroom or computer lab with whiteboard, projector, and math tools | 1 Room    | Shared             |
| 6.    | <b>Infrastructure</b>        | High-speed internet for accessing online simulations and graphing tools       | 1 Setup   | Shared             |
| 7.    | <b>Tools &amp; Equipment</b> | Computers with math software installed                                        | 25        | 1:1                |
| 8.    |                              | Scientific/Graphing calculators                                               | 15        | 1:1.6              |
| 9.    |                              | Rulers, protractors, and compasses for Venn diagrams and graphs               | 25 Sets   | 1:1                |
| 10.   |                              | Whiteboard compasses, markers, visual aids                                    | 1 Set     | Shared             |

|     |                             |                                                                 |         |       |
|-----|-----------------------------|-----------------------------------------------------------------|---------|-------|
| 11. | <b>Consumable Materials</b> | Graph papers, notebooks, pens, printed quizzes, and assignments | 25 Sets | 1:1   |
| 12. |                             | Chart papers for statistics visualizations                      | 10      | 1:2.5 |

## **MODULE VI**

## COMMUNICATION SKILLS

**ISCED UNIT CODE:** 0031 441 01A

**TVET CDACC UNIT CODE:** ICT/CU/ICTA/BC/01/5/MA

**Duration of Unit:** 40 hours

### Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply Communication Skills

### Unit Description

This unit covers the competencies required to apply communication skills. It involves the ability to: apply communication channels, written, non-verbal, oral, and group communication skills.

### Summary of Learning Outcomes

| Learning outcomes                      | Duration (hours) |
|----------------------------------------|------------------|
| 1. Apply communication channels.       | 5                |
| 2. Apply written communication skills. | 10               |
| 3. Apply non-verbal skills.            | 10               |
| 4. Apply oral communication skills.    | 5                |
| 5. Apply group communication skills.   | 10               |
| <b>TOTAL</b>                           | <b>40</b>        |

## Learning Outcomes, Content, and Suggested Assessment Methods

| Learning Outcome                      | Content                                                                                                                                                                                                                                                                                                                                                    | Suggested Assessment Methods                                                                                                                                                                                            |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Apply communication channels       | 1.1 Communication process<br>1.2.1 Principles of effective communication<br>1.2 Channels/medium/modes of communication<br>1.2.2 Factors to consider<br>1.2.3 when selecting a channel of communication<br>1.2.4 Barriers to effective communication<br>1.3 Flow/patterns of communication<br>1.3.1 Sources of information<br>1.3.2 Organizational policies | <ul style="list-style-type: none"> <li>• Oral questions</li> <li>• Written assessment</li> <li>• Observation</li> <li>• Portfolio of Evidence</li> <li>• Practical assessment</li> <li>• Third party report</li> </ul>  |
| 2. Apply written communication skills | 2.1 Types of written communication<br>2.2 Elements of communication<br>2.3 Organization requirements for written communication                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>• Oral assessment</li> <li>• Written assessment</li> <li>• Observation</li> <li>• Portfolio of Evidence</li> <li>• Practical assessment</li> <li>• Third party report</li> </ul> |



|                                          |                                                                                                                                                                                                                            |                                                                                                                                                                                                                         |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3. Apply non-verbal communication skills | 3.1 Utilize body language and gestures<br>3.2 Apply body posture<br>3.3 Apply workplace dressing code                                                                                                                      | <ul style="list-style-type: none"> <li>• Oral assessment</li> <li>• Written assessment</li> <li>• Observation</li> <li>• Portfolio of Evidence</li> <li>• Practical assessment</li> <li>• Third party report</li> </ul> |
| 4. Apply oral communication skills       | 4.1 Types of oral communication pathways<br>4.2 Effective questioning techniques<br>4.3 Workplace etiquette<br>4.4 Active listening                                                                                        | <ul style="list-style-type: none"> <li>• Oral assessment</li> <li>• Written assessment</li> <li>• Observation</li> <li>• Portfolio of Evidence</li> <li>• Practical assessment</li> <li>• Third party report</li> </ul> |
| 5. Apply group discussion skills         | 5.1 Establishing rapport<br>5.2 Facilitating resolution of issues<br>5.3 Developing action plans<br>5.4 Group organization techniques<br>5.5 Turn-taking techniques<br>5.6 Conflict resolution techniques<br>5.7 Team-work | <ul style="list-style-type: none"> <li>• Oral assessment</li> <li>• Written assessment</li> <li>• Observation</li> <li>• Portfolio of Evidence</li> <li>• Practical assessment</li> </ul>                               |

### Suggested Methods of Instruction

- Roleplaying
- Simulation
- Field trips
- Viewing of related videos
- Demonstrations
- Online Training

- Group discussions.
- Instructor led facilitation using active learning strategies.

#### Recommended Resources for 25 trainees

| S/No.    | Category/Item                                   | Description / Specifications                            | Quantity    | Recommended Ratio (Trainee: Item) |
|----------|-------------------------------------------------|---------------------------------------------------------|-------------|-----------------------------------|
| <b>A</b> | <b>Learning Materials</b>                       |                                                         |             |                                   |
| 1        | Textbooks                                       | Relevant to course content                              | 5 pcs       | 5:1                               |
| 2        | PowerPoint Presentations                        | Training slides for instructional delivery              | For trainer | N/A                               |
| 3        | Whiteboard Markers (Assorted)                   | Multi-colour, dry erase markers                         | 2 packets   | For trainer                       |
| 4        | e-Didactics                                     | Digital teaching content                                | For trainer | N/A                               |
| 5        | Flashcards                                      | Used for quizzes, terms, and concept reinforcement      | 25 sets     | 1:1 (or as required)              |
| 6        | Flip Charts                                     | For group work, brainstorming, and visual presentations | 2 pads      | 1:13                              |
| 7        | Whiteboard                                      | For instructional use                                   | 1 unit      | Shared                            |
| <b>B</b> | <b>Learning Facilities &amp; Infrastructure</b> |                                                         |             |                                   |
| 8        | Lecture/Theory Room                             | With furniture, ventilation, power points               | 1 room      | 25:1                              |
| <b>C</b> | <b>Consumable Materials</b>                     |                                                         |             |                                   |
| 9        | Printing Papers                                 | A4 papers for handouts, assessments                     | 1 ream      | 1:20                              |

|          |                            |                                                          |              |        |
|----------|----------------------------|----------------------------------------------------------|--------------|--------|
| 10       | Toners                     | Compatible with printers in use                          | 2 pcs        | 13:1   |
| 11       | Internet                   | Broadband or Wi-Fi for research and online tools         | 1 connection | Shared |
| <b>D</b> | <b>Tools and Equipment</b> |                                                          |              |        |
| 12       | Projectors                 | For displaying teaching materials and multimedia content | 1 unit       | 25:1   |
| 13       | Printers                   | For printing training materials and trainee documents    | 4 units      | 6:1    |
| 14       | Computers / Smartphones    | For simulations, research, online assessments            | 25 units     | 1:1    |

## INFORMATION SYSTEMS DEVELOPMENT

**ISCED UNIT CODE:** 0613 554 11A

**UNIT CODE:** ICT/CU/CS/CR/06/6/MA

### Relationship to Occupational Standards

This unit addresses the unit of competency: Develop Information System

**Duration of Unit:** 200 hours

### Unit Description

This unit covers the competencies required to develop an information system. It involves understanding fundamentals of information systems, applying security measures to data, hardware, software in automated environment, understanding the software development process, demonstrating human computer interaction principles, understanding the VB.net programming environment and developing and testing a VB.NET application.

### Summary of Learning Outcomes

| Learning outcomes                                                         | Duration (hours) |
|---------------------------------------------------------------------------|------------------|
| 1. Fundamentals of Information Systems                                    | 30               |
| 2. Security measures to data, hardware, software in automated environment | 20               |
| 3. Software Development Process                                           | 30               |
| 4. Human Computer Interaction Principles                                  | 20               |
| 5. VB.NET programming environment                                         | 50               |
| 6. Develop and test a VB.NET application                                  | 50               |
| <b>TOTAL</b>                                                              | <b>200</b>       |

## Learning Outcomes, Content and Suggested Assessment Methods

| Learning Outcome                       | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Suggested Assessment Methods                                                                                             |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 1. Fundamentals of Information Systems | <p>1.1 Information systems</p> <p>1.1.1 Definition</p> <p>1.1.2 Components</p> <p>1.2 Types of information systems</p> <p>1.2.1 Transaction Processing Systems</p> <p>1.2.2 Management Information Systems</p> <p>1.2.3 Decision Support Systems</p> <p>1.2.4 Executive Information Systems</p> <p>1.2.5 Office Automation Systems</p> <p>1.3 Emerging trends in information systems</p> <p>1.4 Recommendation of information systems for different scenarios</p> <p>1.5 Information system security</p> <p>1.5.1 Definition</p> <p>1.5.2 Information security management system</p> <p>1.5.3 Tools for information system security</p> <p>1.5.4 Firewalls</p> <p>1.5.5 Virtual private networks</p> <p>1.6 Mobile security</p> <p>1.6.1 Geolocation software</p> <p>1.6.2 Remote data removal software</p> | <ul style="list-style-type: none"> <li>• Oral questioning</li> <li>• Written tests</li> <li>• Practical tests</li> </ul> |

|                                                                           |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                             |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                           | 1.7 Web security<br>1.7.1 Cyber security<br>1.7.2 Technologies<br>1.7.3 Web threats<br>1.7.4 Defence strategies                                                                                                                                                                                                                                  |                                                                                                                                             |
| 2. Security measures to data, hardware, software in automated environment | 2.1 Data security and control<br>2.2 Security threats and control measures<br>2.3 Types of computer crimes<br>2.4 Detection and protection against computer crimes<br>2.5 Laws governing protection of ICT                                                                                                                                       | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Oral questioning</li> <li>• Project</li> <li>• Practical tests</li> </ul> |
| 3. Software Development Process                                           | 3.1 Software Development Life Cycle<br>3.2 Software Development Methodologies<br>3.2.1 Waterfall<br>3.2.2 Spiral<br>3.2.3 Rapid Application Development<br>3.2.4 Agile Development<br>3.3 Modelling techniques<br>3.3.1 Data Flow Diagrams<br>3.3.2 Entity Relation Diagrams<br>3.3.3 UML diagrams<br>3.4 Creation of models for given scenarios | <ul style="list-style-type: none"> <li>• Written tests</li> <li>• Oral questioning</li> <li>• Practical tests</li> </ul>                    |
| 4. Human Computer Interaction Principles                                  | 4.1 Human Computer Interaction<br>4.1.1 Definition<br>4.1.2 Role of interaction design<br>4.1.3 Interaction styles<br>4.1.4 Interaction elements<br>4.1.5 Mistakes in interaction design                                                                                                                                                         | <ul style="list-style-type: none"> <li>• Practical</li> <li>• Oral questioning</li> <li>• Observation</li> <li>• Written tests</li> </ul>   |

|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                    |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                                                      | <p>4.2 Interface design principles</p> <p>4.3 Prescribing interaction choices and recognition of interaction flaws</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                    |
| <p>5. VB.NET programming environment</p>             | <p>5.1 The .Net framework</p> <p>5.1.1 Applications supported</p> <p>5.1.2 Components of the .Net framework</p> <p>5.2 Installation of Visual Studio</p> <p>5.3 Features of VB.Net</p> <p>5.4 The Integrated Development Environment (IDE)</p> <p>5.4.1 Definition of IDE</p> <p>5.4.2 Parts of VB.Net IDE</p> <p>5.5 VB.Net program structure</p> <p>5.5.1 VB.NET syntax</p> <p>5.5.2 Namespace declaration</p> <p>5.5.3 Class or module</p> <p>5.5.4 Procedures</p> <p>5.5.5 Data types, variables, constants</p> <p>5.5.6 The Main procedure</p> <p>5.5.7 Statements and Expressions (Variable declarations, operations, control statements)</p> <p>5.5.8 Comments</p> <p>5.6 Creating aVB.Net project</p> <p>5.6.1 Saving Forms and Project</p> <p>5.6.2 Compiling a Project</p> | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Oral tests</li> <li>• Written tests</li> </ul> |
| <p>6. VB.NET application development and testing</p> | <p>6.1 Basic VB.Net Controls</p> <p>6.1.1 test Controls and their purpose</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• Practical tests</li> <li>• Oral tests</li> </ul>                          |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
|  | <p>6.1.2 Standard naming conventions for controls</p> <p>6.2 Elements of a control</p> <p>6.2.1 Properties</p> <p>6.2.2 Methods</p> <p>6.2.3 Events</p> <p>6.3 Demonstrating Properties, Methods and Events</p> <p>6.3.1 Properties for basic controls</p> <p>6.3.2 Setting properties at design time and run time</p> <p>6.3.3 Methods for basic controls</p> <p>6.3.4 Events for basic controls</p> <p>6.4 Demonstrating event handling</p> <p>6.4.1 Mouse events</p> <p>6.4.2 Keyboard events</p> <p>6.5 Designing VB.NET form using HCI principles</p> <p>6.6 Connection of VB.Net applications to a database</p> <p>6.6.1 ADO.Net object model</p> <p>6.6.2 Demonstrating Database connection using the Data Provider</p> <p>6.6.3 Demonstrating creation of tables using Dataset components</p> <p>6.7 Deployment of VB.NET VB.Net applications</p> <p>6.7.1 Purpose deployment</p> <p>6.7.2 Demonstrating deployment steps</p> | <ul style="list-style-type: none"> <li>• Written tests</li> </ul> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|



## Suggested Methods of Instruction

- Presentations and practical demonstrations by trainer;
- Guided learner activities and research to develop underpinning knowledge;
- Supervised practical assignments and projects;
- Visiting expert from the ICT sector;
- Industrial visits

## Recommended Resources for 25 trainees

| S/No. | Category                     | Item Description / Specifications                    | Quantity  | Recommended Ratio (Trainee: Item) |
|-------|------------------------------|------------------------------------------------------|-----------|-----------------------------------|
| 1.    | <b>Learning Materials</b>    | Textbooks                                            | 25        | 1:1                               |
| 2.    |                              | Printed handouts                                     | 25 Sets   | 1:1                               |
| 3.    |                              | User guides for Visual Studio and VB.NET programming | 25        | 1:1                               |
| 4.    |                              | Access to e-books, online platforms                  | 25 Logins | 1:1                               |
| 5.    | <b>Learning Facilities</b>   | ICT Lab with desks, chairs, projector, whiteboard    | 1 Room    | Shared                            |
| 6.    | <b>Infrastructure</b>        | Internet access                                      | 1 Setup   | Shared                            |
| 7.    |                              | Power backup system                                  | 1 System  | Shared                            |
| 8.    | <b>Tools &amp; Equipment</b> | Computers with Visual Studio & SQL Server installed  | 25        | 1:1                               |
| 9.    |                              | External storage devices                             | 25        | 1:1                               |
| 10.   |                              | Visual Studio Community Edition                      | Installed | N/A                               |
| 11.   |                              | Antivirus/firewall                                   | Installed | N/A                               |
| 12.   | <b>Consumable Materials</b>  | Notebooks, pens, printed worksheets                  | 25 Sets   | 1:1                               |
| 13.   |                              | Project submission sheets, software license logs     | 25 Sets   | 1:1                               |

## ARTIFICIAL INTELLIGENCE CONCEPTS

**ISCED UNIT CODE:** 0613 554 12A

**UNIT CODE:** ICT/CU/CS/CC/04/6/MA

### Relationship to Occupational Standards

This unit addresses the unit of competency: Implement Artificial Intelligence Concepts

**Duration of Unit:** 180 hours

### Unit Description

This unit covers the competencies required to understand artificial intelligence. It involves understanding fundamentals of Artificial Intelligence, understanding problem solving techniques, understanding Python programming environment and developing Artificial Intelligence programs using Python.

### Summary of Learning Outcomes

| Learning outcomes                                | Duration (hours) |
|--------------------------------------------------|------------------|
| 1. Fundamentals of Artificial Intelligence       | 30               |
| 2. Problem solving techniques                    | 40               |
| 3. Python programming environment                | 60               |
| 4. Artificial Intelligence programs using python | 50               |
| <b>TOTAL</b>                                     | <b>180</b>       |

### Learning Outcomes, Content and Suggested Assessment Methods

| Learning Outcome                       | Content                                                                                                                          | Suggested Assessment Methods                                                                                   |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 1. Concepts of Artificial Intelligence | 1.1 Fundamentals Artificial Intelligence<br>1.2 History of Artificial Intelligence<br>1.3 Foundations of Artificial Intelligence | <ul style="list-style-type: none"><li>• Oral tests</li><li>• Written tests</li><li>• Practical tests</li></ul> |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
|                               | 1.3.1 Mathematics<br>1.3.2 Economics<br>1.3.3 Decision Theory<br>1.3.4 Neurology<br>1.3.5 Engineering<br>1.3.6 Psychology<br>1.3.7 Computer Networking<br>1.4 Applications of Artificial Intelligence<br>1.4.1 Expert systems<br>1.4.2 Machine Learning<br>1.4.3 Natural Language Processing<br>1.4.4 Gaming<br>1.4.5 Artificial Neural Networks<br>1.4.6 Computer Vision<br>1.5 Intelligence agents<br>1.6 Recognising Artificial Intelligence applications in real life |                                                                                                                                |
| 2. Problem solving techniques | 2.1 Logical operators<br>2.1.1 AND<br>2.1.2 OR<br>2.1.3 NOT<br>2.2 Propositional Logic and Predicate logic<br>2.3 Types of inferencing<br>2.3.1 Single Inferencing<br>2.3.2 Multiple inferencing<br>2.3.3 Case based reasoning<br>2.4 Machine Learning<br>2.5 Types of Machine Learning                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>• Oral tests</li> <li>• Written tests</li> <li>• Practical tests</li> <li>• </li> </ul> |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                    |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                                   | 2.5.1 Supervised Machine Learning<br>2.5.2 Unsupervised Machine Learning<br>2.6 Recognising applications of different types of inferencing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                    |
| 3. Python programming environment | 3.1 Installation of Python<br>3.1.1 Downloading Python Set Up<br>3.1.2 Running Python Set Up<br>3.2 Python syntax<br>3.2.1 The Zen of Python<br>3.2.2 Python Enhancement Proposals 8 (PEP 8)<br>3.2.3 Variable declaration.<br>3.2.4 Commenting<br>3.3 Python data types<br>3.3.1 Integer<br>3.3.2 Float<br>3.3.3 Boolean<br>3.3.4 Set<br>3.3.5 Dictionary<br>3.3.6 Tuple<br>3.3.7 List<br>3.3.8 String<br>3.4 Control structures in Python<br>3.4.1 Selection<br>3.4.2 Looping<br>3.5 Functions in Python<br>3.5.1 Built-in functions<br>3.5.2 User defined functions<br>3.5.3 Lambda functions<br>3.6 Object Oriented Python | <ul style="list-style-type: none"> <li>• Oral tests</li> <li>• Written tests</li> <li>• Practical tests</li> </ul> |

|                                                              |                                                                                                                                                                                                                                                                                                                                        |                                                                                                                    |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                                                              | 3.6.1 Creation of classes<br>3.6.2 Class variables<br>3.6.3 Class methods<br>3.7 Scientific Modules in Python<br>3.7.1 Pandas<br>3.7.2 Numpy<br>3.7.3 Matplotlib<br>3.8 Creation of applications using Scientific Modules                                                                                                              |                                                                                                                    |
| 4. Artificial Intelligence programs development using python | 4.1 Sci-Kit Learn<br>4.2 Machine Learning with K-Nearest Neighbours<br>4.2.1 Mathematics behind K-Nearest Neighbours<br>4.2.2 Making Predictions with K-Nearest Neighbours<br>4.3 Machine Learning with Naïve Bayes Algorithm<br>4.3.1 Mathematics behind Naïve Bayes Algorithm<br>4.3.2 Making predictions with Naïve Bayes Algorithm | <ul style="list-style-type: none"> <li>• Oral tests</li> <li>• Written tests</li> <li>• Practical tests</li> </ul> |

### Suggested Methods of Instruction

- Presentations and practical demonstrations by trainer;
- Guided learner activities and research to develop underpinning knowledge;
- Supervised practical assignments and projects
- Visiting lecturer/trainer from the Computer Science sector;
- Industrial visits.

### Recommended Resources for 25 trainees

| S/No. | Category                   | Item Description / Specifications                        | Quantity     | Recommended Ratio (Trainee:Item) |
|-------|----------------------------|----------------------------------------------------------|--------------|----------------------------------|
| 1.    | <b>Learning Materials</b>  | Textbooks                                                | 25 Books     | 1:1                              |
| 2.    |                            | Handouts/slides                                          | 25 Sets      | 1:1                              |
| 3.    |                            | Online course subscriptions for AI and Python            | 25 Licenses  | 1:1                              |
| 4.    |                            | Python documentation and cheat sheets (print or digital) | 25 Copies    | 1:1                              |
| 5.    | <b>Learning Facilities</b> | Classroom / ICT Lab with whiteboard, projector, speakers | 1 Room       | Shared                           |
| 6.    |                            | Internet-enabled environment                             | 1 Setup      | Shared                           |
| 7.    | <b>Infrastructure</b>      | High-speed internet connection                           | 1 Connection | Shared                           |
| 8.    |                            | Power backup                                             | 1 System     | Shared                           |

|     |                              |                                                       |            |     |
|-----|------------------------------|-------------------------------------------------------|------------|-----|
| 9.  | <b>Tools &amp; Equipment</b> | Laptops or desktop computers (Python 3.9+ compatible) | 25 Units   | 1:1 |
| 10. |                              | Pre-installed Python                                  | Installed  | N/A |
| 11. |                              | Libraries                                             | Installed  | N/A |
| 12. |                              | External storage                                      | 25 Devices | 1:1 |
| 13. | <b>Consumable Materials</b>  | Notebooks, pens,                                      | 25 Sets    | 1:1 |
| 14. |                              | Assessment forms, quizzes, worksheets                 | 25 Sets    | 1:1 |